United States Patent 12402564
Kind Code B2
Date of Patent September 02, 2025
Inventor(s) Dahl; Andrew et al.

Over-center latch with mount, base plate, lever, and center pin for combine concaves

Abstract

An over-center latch for a concave combine comprises a mount directly attached to a concave combine or an accessory for a concave combine, a base plate, a lever with a loop at one end that is configured to attach to a catch, and a center pin that allows the lever to rotate with respect to a handle. Either the handle or the lever can be pulled into a vertical position.

Inventors: **Dahl; Andrew (Lynn Center, IL), Calmer; Marion (Alpha, IL), Wallerstedt; Daniel (Rio, IL)**

Applicant: **CALMER HOLDING COMPANY, LLC** (Lynn Center, IL)

Family ID: **95562678**

Assignee: **CALMER HOLDING COMPANY, LLC (Lynn Center, IL)**

Appl. No.: **19/012532**

Filed: **January 07, 2025**

Prior Publication Data

Document Identifier	Publication Date
US 20250143221 A1	May. 08, 2025

Related U.S. Application Data

continuation parent-doc US 29868301 20221129 US D1058613 child-doc US 18676197
continuation-in-part parent-doc US 18676197 20240528 US 12225855 child-doc US 19012532
continuation-in-part parent-doc US 18520943 20231128 PENDING child-doc US 19012532
us-provisional-application US 63579718 20230830
us-provisional-application US 63488584 20230306

Publication Classification

Int. Cl.: **A01F12/16** (20060101); **A01D41/12** (20060101); **A01F12/26** (20060101); **E05C19/14** (20060101)

U.S. Cl.:

CPC **A01F12/16** (20130101); **A01D41/1261** (20130101); **A01F12/26** (20130101); **E05C19/14** (20130101);

Field of Classification Search

CPC: E05C (19/14); A01F (12/16); A01F (12/26); A01D (41/1261)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1002901	12/1910	Christman	267/74	B60P 7/0838
1840037	12/1931	Johansson	N/A	N/A
3162419	12/1963	Blasingame	403/325	F16M 7/00
3519298	12/1969	Gley	292/108	E05C 19/14
3568682	12/1970	Knapp et al.	N/A	N/A
4004404	12/1976	Rowland-Hill et al.	N/A	N/A
4218081	12/1979	Johnson	292/66	E05C 19/14
D257218	12/1979	Eriksson	D8/336	N/A
4407493	12/1982	Okolischan	N/A	N/A
4422218	12/1982	Brasseux	N/A	N/A
4493133	12/1984	Nilsson	24/68CD	E05C 19/14
4711075	12/1986	Strong	N/A	N/A
4711252	12/1986	Bernhardt et al.	N/A	N/A
D329686	12/1991	Mathison	D23/249	N/A
D329687	12/1991	Benson	D23/249	N/A
5165148	12/1991	Fleischer	292/113	B25B 5/12
5478125	12/1994	Gromotka	292/113	E05C 19/14
5489239	12/1995	Matousek	460/109	A01F 12/28
5667261	12/1996	Weinerman et al.	N/A	N/A
5919087	12/1998	Strong	N/A	N/A
6179350	12/2000	Ely	292/285	E05C 19/14
6374465	12/2001	Dykstra	N/A	N/A
6561556	12/2002	Fuchs	292/250	B65D 45/345
7001268	12/2005	Puryk	460/59	A01F 12/185
7017955	12/2005	Chiang	292/113	E05C 19/14
7059961	12/2005	Schenk	460/59	A01D 75/182

7118475	12/2005	Schenk	N/A	N/A
7156429	12/2006	Eriksson	292/113	F16B 2/185
7213848	12/2006	Hagan	24/494	E05C 19/14
7285043	12/2006	Foster et al.	N/A	N/A
D624939	12/2009	Flickinger et al.	N/A	N/A
8454416	12/2012	Estes	N/A	N/A
D687659	12/2012	MacKenzie	N/A	N/A
8561973	12/2012	Martin	269/201	E05C 19/14
8690652	12/2013	Estes	N/A	N/A
8721411	12/2013	Reiger et al.	N/A	N/A
D727134	12/2014	Yamamoto et al.	N/A	N/A
9080353	12/2014	Washio	N/A	N/A
D735772	12/2014	Ricketts et al.	N/A	N/A
9504204	12/2015	Kile	N/A	N/A
9723791	12/2016	Kile	N/A	N/A
9763390	12/2016	Matway et al.	N/A	N/A
9913431	12/2017	Van Hullebusch	N/A	N/A
10045487	12/2017	Robertson	N/A	N/A
10143141	12/2017	De Witte	N/A	N/A
10405494	12/2018	Regier	N/A	N/A
10440893	12/2018	Kile	N/A	N/A
10602667	12/2019	Ritter	N/A	N/A
10779474	12/2019	Ritter	N/A	N/A
10849275	12/2019	Estes	N/A	N/A
10869427	12/2019	Robertson	N/A	N/A
10905050	12/2020	Robertson	N/A	N/A
11116136	12/2020	Robertson	N/A	N/A
11122743	12/2020	Robertson	N/A	N/A
11317566	12/2021	Koudela	N/A	N/A
D954759	12/2021	Robertson	N/A	N/A
11375667	12/2021	Broholm et al.	N/A	N/A
11497171	12/2021	Theisen et al.	N/A	N/A
11877538	12/2023	Calmer et al.	N/A	N/A
2007/0178951	12/2006	Voss et al.	N/A	N/A
2015/0250101	12/2014	Kile	N/A	N/A
2015/0313085	12/2014	Coppinger	N/A	N/A
2016/0345499	12/2015	Van Hullebusch	N/A	N/A
2017/0105351	12/2016	Matway	N/A	N/A
2018/0084728	12/2017	Reiger	N/A	N/A
2018/0103588	12/2017	Ritter	N/A	N/A
2018/0352746	12/2017	Ritter et al.	N/A	N/A
2018/0359926	12/2017	Estes	N/A	N/A
2019/0037773	12/2018	Theisen et al.	N/A	N/A
2019/0159403	12/2018	Haar et al.	N/A	N/A
2019/0166767	12/2018	Robertson	N/A	N/A
2019/0166768	12/2018	Robertson	N/A	N/A
2019/0200529	12/2018	Robertson	N/A	N/A
2020/0196530	12/2019	Van Hullebusch et al.	N/A	N/A
2021/0015048	12/2020	Kile	N/A	N/A

2021/0105946	12/2020	Robertson	N/A	N/A
2022/0194149	12/2021	Choe	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4431372	12/1994	DE	E05C 19/14
590033	12/1924	FR	N/A
2726161	12/1995	FR	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion in PCT/US2023/081250, 13 pages, mailed Apr. 18, 2024. cited by applicant
John Deere, Product Catalog, www.JohnDeere.com, 128 pages, Spring 2021. cited by applicant

---

Primary Examiner: Fabian-Kovacs; Arpad

Attorney, Agent or Firm: HAMILTON IP LAW

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part (CIP) application of U.S. Ser. No. 18/676,197, titled “KITS AND COMPONENTS FOR REGULATING THE FLOW OF MATERIAL OTHER THAN GRAIN THROUGH COMMON CONCAVES WHILE SIMULTANEOUSLY EQUALIZING THE FLOW OF GRAIN THROUGH COMMON CONCAVES”, filed May 28, 2024, which is a continuation-in-part (CIP) application of U.S. Ser. No. 18/520,943, titled “METHODS OF REGULATING THE FLOW OF MATERIAL OTHER THAN GRAIN THROUGH COMMON CONCAVES WHILE SIMULTANEOUSLY EQUALIZING THE FLOW OF GRAIN THROUGH COMMON CONCAVES”, filed Nov. 28, 2023, which claims priority under 35 U.S.C. § 119 to provisional patent applications U.S. Ser. No. 63/488,584, titled “SERIES OF ACTUATABLE MATERIAL OTHER THAN GRAIN LIMITING INSERTS FOR COMMON CONCAVE”, filed Mar. 6, 2023; and U.S. Ser. No. 63/579,718, titled “METHODS OF REGULATING THE FLOW OF MATERIAL OTHER THAN GRAIN THROUGH COMMON CONCAVES WHILE SIMULTANEOUSLY EQUALIZING THE FLOW OF GRAIN THROUGH COMMON CONCAVES”, filed Aug. 30, 2023; and further claims priority under 35 U.S.C. § 120 to design patent application U.S. Ser. No. 29/868,301, titled “LIMITER PLATES”, filed Nov. 29, 2022, now U.S. Design Pat. No. D1,058,613, issued Jan. 21, 2025. These applications and provisional patent applications are herein incorporated by reference in their entireties, including without limitation, the specification, claims, and abstract, as well as any figures, tables, appendices, or drawings thereof.

TECHNICAL FIELD
(1) The present disclosure relates generally to an apparatus and method applicable to the harvest of agricultural crops including a variety of small grain crops. A persistent problem presented to farmers in the threshing operation is the separation of the grain from the plant's stems, leaves, weeds, volunteer plants, and other trash or foreign materials, all of which is referred to as material(s) other than grain (MOG), which are all introduced to the combine in the harvest operation. Critical to maximizing the capture of the grain in harvest and the return obtained by the farmer is the robust, yet efficient separation of the grain from MOG. In particular, but not exclusively, the present disclosure relates to a series of MOG limiting devices that attach to and function in cooperation with a concave positioned within a combine.

BACKGROUND

(2) The background description provided herein gives context for the present disclosure. Work of the presently named inventors, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art.

(3) Historically, a number of issues confront the farmer in conjunction with the harvest of large grains and small grains. In one regard, a problem presented to the farmer in harvest is the successful separation of the grain from the MOG which includes the application of abrasing forces upon the crop by acting upon itself as well as the metallic threshing components in the rotor chamber of the combine. Once the grain is separated, there is a need to rapidly direct the grain out of the rotary chamber of the combine while retaining the MOG within the chamber so as to maintain a low chaff load and not overwhelm the grain processing and cleaning system of the combine with MOG thus allowing for grain loss, generally referred to as sieve loss and the introduction of MOG into the grain bin resulting in the application of a dockage penalty to the farmer due to the foreign matter (FM), also referred to as foreign material, in the grain delivered for storage and sale.

(4) In the harvest of large grain like corn, it can be difficult to divert the threshed kernels out of the rotary chamber, leaving the kernels to be acted upon by the continued threshing forces which may result in the fracture or milling of the kernel to create fines. The net result is loss of yield to the farmer. To prevent this damage and yield loss, large openings may be provided in the concaves which allows for expedited removal of the kernels, but also provides a ready pathway for large volumes of MOG to enter the cleaning system.

(5) In the harvest of small grains, the issue presented to the farmer is to retain the crop within the threshing chamber for a sufficiently long time to allow for complete separation of the grain from the pod or other plant covering.

(6) A notable problem with soybeans is an evolving moisture difference in the crop. In past years soybean harvest was made easier by a similar moisture content between soybean pods and stems. In recent years, soybean header widths have grown from 10 feet to 50 feet, and the headers are used on combines having dynamically increased horsepower and capacity. Most recently, the introduction of fungicides which operate to extend plant life and genetic improvements in soybean varieties contribute to difficulties in that the combine now encounters green pods, wet pods, and dry normal pods all in the same swath as well as greener stems then encountered in years past. This creates a wide range of moisture content in the material being processed within the rotary chamber of the combine. As a part of this issue, farmers no longer wait for moisture content in the soybeans to become more homogeneous given the increased number of acres to be harvested prior to the onset of colder and wetter weather. Thus, the threshing operation is faced with the requirement to process green stems with dry stems and green pods with dry pods. As a result, there is a need for increases in threshing power while simultaneously reducing the amount of MOG that passes into the cleaning system. Finally, and notably, the green stems increases the problems of plugging in both the concaves as well as the separation plates—all to the effect of increasing sieve loss.

(7) In this effort to achieve a balance between the separation and capture of the grain while, at the same time, retaining the MOG within the rotary chamber until it is ejected at the back end onto the field historically has resulted in around the adjustment of the combine. Also, the configuration of the threshing concaves and separation grates played an important role in the threshing of the grain and removal from the rotor chamber of the combine, however, it was and remains impractical for the farmer to change out the concaves when transitioning from one crop to another for harvest. Cover or filler plates have been available to the farmer for use with the threshing concaves, but these have been most difficult to install and, further, operated to dam-up or restrict the entire open area on each half of the concaves. This configuration operated to retain the grain within the rotor chamber with resultant damage.

(8) Given the higher volume of plant materials processed by the combine in harvest in combination

with the differing moisture values of the plant material, there exists a need in the art for an apparatus which allows for case of installation combined with a refinement of operation and adjustability so as to produce the maximum beneficial results for the farmer, including but not limited to reducing damage and yield loss to grain while also reducing MOG found in an end product.

SUMMARY

(9) A list of additional problems known in the art follow.

(10) Large wire concaves have been used successfully for many years in position #1 in older combines that were smaller and designed to handle lower corn yields than commonly grown today. In today's high-yielding corn crops harvested in higher-moisture conditions newer combines have more horsepower and bigger appetites. The advancements have overwhelmed the capacity of the concave designs that worked sufficiently in the past. Large wire concave limitations retain shelled kernels in the rotor chamber long enough to be damaged. The low pass-through rate manifests the problems of rotor loss, broken cobs, and damaged grain in the sample. Combine models known in the art need to improve their capacities. Large wire concaves used in position #1 need to increase their concave open areas in an optimized way, all while saving investment dollars in newer equipment.

(11) Concaves known in the art cause broken or splintered cobs in the grain tank because they unnecessarily employ large wire concaves used in positions #2 and #3, thereby causing unnecessary cob breakage. When cobs are broken or splintered into smaller pieces, they fall into the cleaning area and potentially end up in the grain tank.

(12) Many farmers make the mistake of not considering and/or not utilizing cover plates in concave position #1 when harvesting soybeans. When harvesting soybeans, most concaves used for corn harvest have openings too large to retain pods or small grains long enough for them to thresh fully (see, e.g., FIG. **46** including a pod). Instead, when they enter the threshing chamber, the unthreshed pods fall directly through these large openings onto the auger bed. These unthreshed pods then move into and overload the cleaning system and will need to be separated and sent back a second time to thresh. This process is commonly referred to as “tailings.” The tailings also contain clean threshed soybeans, which are unnecessarily re-threshed and can become split. A major problem occurs where pods are mixed with split beans and cannot be differentiated before harvest.

(13) Another major problem in the art occurs where OEM concaves are used for harvesting all crops in all conditions. OEM concaves are either better suited for the harvest of soybeans or maize, for wet conditions or for dry conditions. A middle of the road solution does not work. If the OEM spacing between round bars in round bar concaves is aimed to be used to harvest corn and soybeans, the gaps are too narrow for corn and too wide for harvesting soybeans.

(14) Some modern and/or persisting problems in corn include but are not limited to rotor loss, damaged grain, splintered cobs, and balancing the combination thereof.

(15) Some modern and/or persisting problems in soybeans and other small grains include but are not limited to pods in tank, high tailings, and sieve loss.

(16) The present disclosure sets forth a solution that enhances flow by allowing for the easy installation of inserts when harvesting all crops in all conditions. This is because the solution allows the wider round bar spacing increases flow in corn and uses inserts to narrow the spacing when harvesting all other crops.

(17) The following objects, features, advantages, aspects, and/or embodiments, are not exhaustive and do not limit the overall disclosure. No single embodiment need provide each and every object, feature, or advantage. Any of the objects, features, advantages, aspects, and/or embodiments disclosed herein can be integrated with one another, either in full or in part.

(18) It is a primary object, feature, and/or advantage of the present disclosure to improve on or overcome the deficiencies in the art. Such deficiencies include but are not limited to a lack of control over the gap (opening) size presented by the concaves and/or separation grates within a

combine which operate to allow the harvested grain to drop out of the rotor chamber and into the cleaning system for movement to the grain bin of the combine. Presently, the control over the gap size is dependent upon removal of a concave or separation grate and substitution of a completely differing one with a differing gap opening size or configuration or, alternatively, simply plugging the opening to effectively prevent anything from passing out of the rotor chamber. One embodiment of the present disclosure controls the gap size to effectively prevent MOG from passing through or plugging the opening while allowing the grain to drop out of the rotor chamber.
(19) It is a further object, feature, and/or advantage of the present disclosure to independently utilize each of the series of MOG limiting devices, also described herein as MOG limiters. In one embodiment the MOG limiter may be approximately one-quarter of the circumference of the rotor or approximate 50% of the length of the concave openings, and such MOG limiter may be referred to as a Quarter Wrap MOG limiter. In another embodiment, the MOG limiter may be of a length to cover all or nearly all of the openings on one or both sides of the concave which limiter may be referred to as a Half Wrap MOG limiter.
(20) The MOG limiters can be selectively attached to one or more of the concaves of a combine, such as the combine shown and described by the inventors in the co-owned, co-pending patent application U.S. Ser. No. 17/899,240, titled THRESHING GRAINS AND LEGUMES UTILIZING CONCAVES WITH ADJUSTABLE OPENINGS, filed Aug. 30, 2022, which is herein incorporate by reference in its entirety. In addition to the selectivity of restriction as to parts of one or more of the concaves, the MOG limiting devices more closely align to the curvature of radius of the concaves and utilize a preset adjustment to position the insert within each gap of the concave so as to control the remaining opening of the gap.
(21) Another embodiment of the MOG limiter allows for an adjustable tensioning connection that not only secures the limiter in place, but also operates to control the positioning of the insert within the gap so as to control remaining gap size existing between the concave crossbars.
(22) It is still yet a further object, feature, and/or advantage of the present disclosure to improve upon the efficiency of the harvest of soybeans or other small grains where concave cover plates have heretofore been used.
(23) It is still yet a further object, feature, and/or advantage of the present disclosure to install a one or more of the MOG limiters which may be placed on either side of the concave or on both sides of the concave. In total, at least four independently actuatable Quarter Wrap MOG limiters may be installed on each concave. In one embodiment a first plurality of MOG limiters are arranged in series and attach to the right side of the concave and arranged in series; and a second plurality of MOG limiters are arranged in series and are attached to a left side of the concave and arranged in series. In one embodiment a first plurality/series of MOG limiters and a second plurality/series of MOG limiters are arranged in parallel. The MOG limiters can approximate one-half of the curvilinear length of a concave referred to herein as Quarter Wrap MOG limiters or MOG limiters, however it is to be appreciated that embodiments exists where the MOG limiters can approximate one third, one quarter, one fifth, etc. of the curvilinear length of the concave, thereby bringing the total number of independently actuatable MOG limiting devices employed on a single concave to six, eight, ten, etc. In various embodiments, each MOG limiter operates to regulate and distribute the flow of MOG in all rotary combines, single or dual rotor designs, passing through the combine in both the rotor chamber and cleaning system.
(24) A first embodiment of a modified concave includes an enhanced side flow rail, a C-clamp end, a first-end round bar drive, a second end plate, a quarter wrap center catch, an enhanced flow center rail, 0.625 inch diameter round bars, a left over-center catch, and a right over-center catch.
(25) A second embodiment of a modified concave includes an enhanced side flow rail, a C-clamp end, a first-end round bar drive, a second end plate, a quarter wrap center catch, an enhanced flow center rail, 0.375 inch rub bars with 0.125 inch prouds, 0.625 inch diameter round bars, a left over-center catch, and a right over-center catch.

(26) A third embodiment of a modified concave includes an enhanced side flow rail, a C-clamp end, a first-end round bar drive, a second end plate, a quarter wrap center catch, an enhanced flow center rail, 0.25 inch flat rub bars, 0.625 inch diameter round bars, a left over-center catch, and a right over-center catch.
(27) A fourth embodiment of a modified concave includes an enhanced side flow rail, a C-clamp end, a first-end round bar drive, a second end plate, a quarter wrap center catch, an enhanced flow center rail, 0.375 inch flat rub bars, 0.625 inch diameter round bars, a left over-center catch, and a right over-center catch.
(28) It is still yet a further object, feature, and/or advantage of the present disclosure to provide a hinged connection in the middle of the concave to grab on to each of the first plurality of MOG limiters and the second plurality of MOG limiters. The hinged connection can be a hook.
(29) It is still yet a further object, feature, and/or advantage of the present disclosure to use an adjustable tensioning connection to secure the MOG limiter in place after rotating the hinged connection to a closed position.
(30) It is still yet a further object, feature, and/or advantage of the present disclosure to be able to install one or more MOG limiting devices which allows for controlled and adjustable restrictions of the concave gaps. When secured in place, the MOG limiter aligns the inserts against one side of the round bars of the concave or may position the inserts somewhere within the gap between the round bars to provide more restricted openings to more beneficially impact the capture of grain and the rejection of MOG.
(31) It is still yet a further object, feature, and/or advantage of the present disclosure to widen the air gap between round bars to provide better solutions for harvest of corn, while simultaneously being able to narrow air gaps for harvest of soybeans. For example, ¼ wrap cover plates and ¼ wrap inserts can be quickly installed and/or interchanged to provide an application specific approach for the harvest of corn or soybeans.
(32) It is still yet a further object, feature, and/or advantage of the present disclosure to uniformly distribute grain and MOG in the threshing and separation chambers of the combine. For example, the MOG limiters and/or cover plates described herein can balance an amount of MOG from a rotor area of the combine to a sieve area (also referred to as the cleaning system) of the combine and allow balance of grain throughput from the rotor area to the sieve area with respect to the left side of the concave to the right side of the concave.
(33) It is still yet a further object, feature, and/or advantage of the present disclosure to thresh most of the grain so that it falls out as quickly as possible so that it goes into the auger bed, which is beneath the rotor. Then, it is desirable to separate and to drop the last grain onto the auger bed before the grain exits the rotor and goes onto the ground as rotor loss. The last of the grain should ideally drop out at the last separating grate.
(34) It is still yet a further object, feature, and/or advantage of the present disclosure to maintain the use of the OEM sieves, which are non-adjustable sieves (large and heavy components).
(35) It is still yet a further object, feature, and/or advantage of the present disclosure to place a highest importance on the choice of combines, MOG limiters, and cover plates for the concaves that are in positions #1-#3.
(36) It is still yet a further object, feature, and/or advantage of the present disclosure to improve harvest of grain such that there is 40% less grain loss (˜1 kernel/sq ft; 2 soybeans per square foot); an increase in ground speed; less damaged grain; and/or less frustration to the operator of the combine.
(37) It is still yet a further object, feature, and/or advantage of the present disclosure to adjust or achieve a proper balance in the expedited separation of grain into the processing and cleaning system of the combine while retaining MOG within the rotary chamber to be passed out the back, which balance may allow for the removal of bottom sieve in the cleaning system which often operates to disrupt air flow blowing chaff out of the combine and away from the pathway moving

grain up to the storage bin.

(38) The present disclosure addresses difficulties presented by the moisture level of the grain being harvested. For example, with corn the use of MOG limiters operates to improve the concave flow-through of small kernel corn in low moisture conditions while preventing the passage of MOG into the cleaning system. In general, as grain moisture in the harvested crop decreases, the usefulness of MOG limiters increases.

(39) The present disclosure addresses difficulties in harvesting small grains including small kernel corn after installing the concaves. For example, it may be necessary to level the rotor after installing the concaves described herein, else the threshing efficiency of the concaves can be diminished. In yet another example, one or more additional inserts and/or cover plates can be added and/or removed at various positions in the threshing and separation chambers (e.g. positions #1-#7) when harvesting tougher crops so as to more fully remove and/or reduce white caps in the sample. In yet another example, the rear separation grates can be rotatably moved into an “up” position, wherein spacers are removed and stored in a closed post.

(40) Cover plates can be used to improve the threshing capability of the rasp bar threshing cylinder while simultaneously capturing additional threshed grain.

(41) The selective use of cover plates versus inserts, including the selective use of same within a single concave, can help balance the flow of MOG and grain through the combine such that it is more equally distributed when passing through various chambers of said combine. This can allow for the better and more efficient harvest a wider variety of crop types without needing to spend large amounts of time installing and uninstalling non-adjustable cover plates, inserts, and the like that at present plague the art.

(42) The cover plate can include an elongated plate body, such as a curvilinear plate, dimensioned to be positioned between two parallel arcuate rails of a threshing concave grate assembly. The curvilinear plate preferably does not span the entire length of said arcuate rails. The curvilinear plate of the cover plate assembly is preferably constructed of a single plate of high strength material such as metal, high strength plastics or composite fabric material. For example, the curvilinear plate can be constructed of steel plating that is permanently bent in an arcuate shape matching the exterior arcuate shape of a threshing concave grate assembly. The curvilinear plate can also be constructed of flexible, high strength materials such as stainless steel or woven composite materials, or alternatively, the cast into a predetermined arcuate shape. It is to be appreciated the material that the elongated body is constructed of may also have a natural porosity and permeability. For example, the elongated body of the cover plate may be constructed of a metal mesh or composite material having organic porosity and permeability characteristics. This can help regulate moisture release from the cover plates should they become wet.

(43) The present disclosure addresses difficulties presented while harvesting in-field. For example, the clearance between the rotor and concaves can be increased and/or decreased during operation of the combine. In yet another example, the operator may increase and/or decrease the rotational speed of the rotor. In yet another example, the operator may alter a velocity of fan speeds. In yet another example, the operator may adjust a distance between the chaffer, sieve(s), concave(s), grate(s), and/or components thereof (e.g., calibrating a position of the louvers that are included in the bottom sieve), etc.

(44) Crops such as rice, wheat, and barley comprise three main parts: the grain—the seeds located at the top; the chaff—the seeds' dry coatings; and the stalk—the overall stem of the plant. Not all three parts are edible. Only the grain can be collected for use in bread and cercal production, which is edible. The other two—chaff and stalk—have to be removed. Threshing refers to the task of separating grain from its attached crop. This grain preparation task comes after the reaping process. Winnowing takes place right after threshing. The process of winnowing frees the grain from the chaff.

(45) The present disclosure addresses difficulties in efficiently threshing grain and those presented

by dirty samples. For example, unthreshed grain can be placed in a return of the combine for re-threshing. In some embodiments, the return runs at least at 50% full for proper re-threshing. Where a dirty sample is concerned, a power shut down may be required so that any grain and chaff in the auger bed and/or top chaffer can be dislodged, and the combine can return to more evenly distribute material across the top chaffer and/or bottom sieve.

(46) The MOG limiters disclosed herein can be used in a wide variety of applications for the harvest of a number of crops including, but not limited to soybeans, wheat, oats, rye, barley, sorghum, flax, sunflowers and canola, alfalfa, clover, and also low yielding corn.

(47) It is preferred the threshing systems, accessories, and configurations disclosed herein be safe, cost effective, and durable. For example, the case of installation and removal allows the farmer to only install and utilize the MOG limiters and cover plates when needed for the specific grain being combined thereby reducing wear. Further, the radial geometry and profile of the MOG limiters and cover plates, and the composition of the steel from which they are constructed, allow for extended use over the years while resisting excessive abrasive wear and/or mechanical failures (e.g. cracking, crumbling, shearing, creeping) due to excessive impacts and/or prolonged exposure to tensile and/or compressive forces which may act upon the MOG limiters and cover plates when positioned within the rotor chamber of an operational combine.

(48) At least one embodiment disclosed herein comprises a distinct aesthetic appearance. Ornamental aspects included in such an embodiment can help capture a consumer's attention and/or identify a source of origin of a product being sold. Said ornamental aspects will not impede functionality of the combine.

(49) Methods can be practiced which facilitate use, manufacture, assembly, maintenance, and repair of MOG limiters and cover plates which accomplish some or all of the previously stated objectives.

(50) The MOG limiters and cover plates may be incorporated independently into the harvest system of a combine, single rotor or dual rotor, or may be included in kits which provide additional components to accomplish some or all of the previously stated objectives. For example, kits that upgrade existing concaves or upgrade existing combines with concaves are disclosed herein. The kits can include pairs of MOG limiters that are intended to span half of the length of the concave and half of the width of the concave, and further, pairs of quarter-wrap cover plates. These kits can be installed on said concaves and/or combines depending on the crop to be harvested or the conditions in which harvest occurs. The components in the kit are easily interchanged. The kits can include various numbers of pairs of MOG limiters and cover plates, depending on the climate in which the kit is expected to be used.

(51) In some combines, it may be advantageous to use a full-length MOG limiter plate which covers one-half of the concave. One instance involves a combine structure wherein the shortened MOG limiter does not work due to connection problems resulting from lack of access from one side of the rotor chamber.

(52) The adjustable use of the present disclosure provides the user the ability to more precisely balance the MOG retention within the rotor chamber and the release of grain from the rotor chamber and, also, to balance the distribution of material released from the rotor chamber onto the sieve for more efficient cleaning of the grain and increased retention of the grain during the harvest process. In areas where a smaller grain such as soybeans is harvested, immediately followed by the harvest of a large grain such as corn, the ability to adjust or easily remove MOG limiters and/or cover plates is important to facilitation of a seamless harvest which allows for optimum use of good weather days. It is also desirable to be able to adjust the size of the gaps and/or openings in the concave without needing to remove the cover plates and/or inserts, to better accommodate a wider range and/or varying in field conditions (e.g., changing moisture conditions).

(53) In some embodiments of some aspects of the present disclosure there includes an over-center latch for attaching a cover plate or MOG limiter to an outer surface of a concave for a combine including a subbase including a pair of ears protruding therefrom, a handle in rotational

communication with the pair of ears at a base of the handle, a center pin positioned between the base of the handle and a distal end of the handle configured to be in rotational communication with the handle, and further there includes a lever attached at a base of the lever to the center pin, wherein the center pin allows the lever to rotate with respect to a handle, and additionally there includes a loop at a distal end of the lever opposite the base of the lever. For this embodiment either the handle or the lever can be pulled into a vertical position orthogonal the subbase.

(54) According to additional aspects of the embodiment, a lock hole protruding from the subbase in a same direction as the pair of ears.

(55) According to additional aspects of the embodiment, a locking mechanism positioned within the lock hole.

(56) According to additional aspects of the embodiment, the locking mechanism is a linchpin.

(57) According to additional aspects of the embodiment, a threaded portion along a length of the lever that extends into the center pin, wherein the center pin includes a threaded section to receive the threaded portion of the lever.

(58) According to additional aspects of the embodiment, a locking nut that can attach the threaded portion of the lever to the center pin.

(59) According to additional aspects of the embodiment, a nylon friction ring positioned within the locking nut that dampens vibration.

(60) According to additional aspects of the embodiment, the handle includes an angled portion on the distal end of the handle to assist with lifting the handle, wherein the angled portion is angled away from the subbase when the handle is positioned substantially parallel to the subbase.

(61) According to additional aspects of the embodiment, an over-center latch assembly including the over-center latch and further comprising a base plate configured to attach to the subbase of the over-center latch.

(62) According to additional aspects of the embodiment, the subbase includes a plurality of apertures and the base plate includes a plurality of holes, wherein the apertures may align with the holes for connecting the base plate to the subbase.

(63) According to additional aspects of the embodiment, threaded rods or screws inserted through the holes and the apertures for connecting the subbase and base plate, wherein the threaded rods or screws include a locking nut with a nylon insert to fasten the threaded rods or screws to the over-center latch assembly.

(64) According to additional aspects of the embodiment, the subbase and the base are a monolithic component via weld such that the pair of ears extend directly from the base plate.

(65) According to additional aspects of the embodiment, the base plate comprises substantially a pentagonal shape.

(66) According to additional aspects of the embodiment, the base plate comprises substantially a rectangular shape.

(67) According to additional aspects of the embodiment, the over-center latch assembly attaches to either of a MOG limiter or a cover plate via the base plate.

(68) According to additional aspects of the embodiment, the base plate includes slots positioned at opposite ends of the base plate and the base plate attaches to the MOG limiter or the cover plate via plug welds in the slots.

(69) According to additional aspects of the embodiment, the over-center latch assembly secures the cover plate or the MOG limiter to an outer surface of a concave.

(70) According to additional aspects of the embodiment, the concave includes catches for the loop to connect to for securing the cover plate or the MOG limiter to the concave.

(71) According to other embodiments of other additional aspects of the present disclosure, a method for securing a MOG limiter or cover plate to a concave via an over-center latch assembly, including positioning a hinge point of the MOG limiter or cover plate within a cross bar of the concave, lifting a handle of the over-center latch such that a lever operatively connected to the

handle extends outwardly away from the MOG limiter or cover plate, placing a loop that is positioned at a distal end of the lever on a catch of the concave, rotating the handle a direction opposite the lifting such that lever retracts back towards the MOG limiter or cover plate, and placing a locking mechanism through a lock hole on the over-center latch assembly after the handle has been rotated past the lock hole.
(72) According to other embodiments of other additional aspects of the present disclosure, a combine including a concave, including a plurality of cover plates and/or MOG limiters positioned on an outer surface of the concave and a plurality of over-center latch assemblies configured to secure the plurality of cover plates and/or MOG limiters to the outer surface of the concave, wherein each of the cover plates and/or MOG limiters includes one of the plurality of over-center latch assemblies secured thereto, wherein each of the plurality of over-center latch assemblies comprises a base plate with a pair of ears extending therefrom; a handle operatively connected to the pair of ears; a center pin positioned along a length of the handle and capable of rotating with relation to the handle; a lever extending from the center pin; and a loop positioned at a distal end of the lever; wherein the cover plates and/or MOG limiters are secured to the concave via insertion of hinge points into a cross bar of the concave, wherein the hinge points are positioned on the cover plate and/or MOG limiter at an end opposite the over-center latch assembly, wherein the cover plate and/or MOG limiter is further secured to the concave via attaching the loops of the over-center latch assemblies to catches positioned at each end of the concave.
(73) These and/or other objects, features, advantages, aspects, and/or embodiments will become apparent to those skilled in the art after reviewing the following brief and detailed descriptions of the drawings. The present disclosure encompasses (a) combinations of disclosed aspects and/or embodiments and/or (b) reasonable modifications not shown or described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Several embodiments in which the present disclosure can be practiced are illustrated and described in detail, wherein like reference characters represent like components throughout the several views. The drawings are presented for exemplary purposes and may not be to scale unless otherwise indicated.
(2) FIG. **1** shows a perspective view of a prior art example of a combine in which embodiments of the present disclosure may be used.
(3) FIG. **2** shows an angled perspective of the underside of a concave with the MOG limiter, according to an embodiment of the present disclosure.
(4) FIG. **3** shows an angled perspective view of the topside of one embodiment of FIG. **2**.
(5) FIG. **4** shows a detailed view of a MOG limiter mounted behind the round bars of a concave.
(6) FIG. **5** shows a bottom plan view similar to that shown in FIG. **2**, including exemplary hardware for attaching the MOG limiter to the concave at the upper and lower connections, according to an embodiment of the present disclosure.
(7) FIG. **6** shows an angled perspective view of a MOG limiter, according to an embodiment of the present disclosure.
(8) FIG. **7** is an angled perspective view showing underside of the embodiment of FIG. **6**.
(9) FIG. **8** is an angled view showing a topside of the MOG limiter including off-center notches, according to an embodiment of the present disclosure.
(10) FIG. **9** is an angled view showing a topside of the MOG limiter including centered notches, according to an embodiment of the present disclosure.
(11) FIG. **10** is a first angled perspective view showing a left-side cover plate that can mount to the concave of a combine.

(12) FIG. **11** is a second angled perspective view of the cover plate of FIG. **10**.
(13) FIG. **12** is a first angled perspective view showing a right-side cover plate that can mount to the concave of a combine.
(14) FIG. **13** is a second angled perspective view of the cover plate of FIG. **12**.
(15) FIG. **14** is a concave, a MOG limiter, and a cover plate combination that includes two right-side cover plates, one left-side cover plate, and a MOG limiter installed on the left-side of the concave.
(16) FIG. **15** is a second angled perspective view of the combination of FIG. **14**.
(17) FIG. **16** is a third angled perspective view of the combination of FIG. **14**.
(18) FIG. **17** is a fourth angled perspective view of the combination of FIG. **14**.
(19) FIG. **18** is a concave, a MOG limiter, and a cover plate combination that includes two left-side cover plates, one right-side cover plate, and a MOG limiter installed on the right-side of the concave.
(20) FIG. **19** is a second angled perspective view of the combination of FIG. **18**.
(21) FIG. **20** is a third angled perspective view of the combination of FIG. **18**.
(22) FIG. **21** is a fourth angled perspective view of the combination of FIG. **18**.
(23) FIG. **22** shows an example configuration that employs a plurality of distinct combinations of concaves with MOG limiters and/or cover plates for the distinct concave position(s) of a combine, including the combination of FIG. **14** shown therein by way of example.
(24) FIG. **23** shows a perspective view of a first embodiment of a modified MOG limiting concave assembly.
(25) FIG. **24** shows a rear elevation view of the first embodiment of the modified MOG limiting concave assembly of FIG. **23**.
(26) FIG. **25** shows a perspective view of a second embodiment of a modified MOG limiting concave assembly.
(27) FIG. **26** shows a rear elevation view of the second embodiment of the modified MOG limiting concave assembly of FIG. **25**.
(28) FIG. **27** shows a perspective view of a third embodiment of a modified MOG limiting concave assembly.
(29) FIG. **28** shows a rear elevation view of the third embodiment of the modified MOG limiting concave assembly of FIG. **27**.
(30) FIG. **29** shows a first perspective view of a fourth embodiment of a modified MOG limiting concave assembly.
(31) FIG. **30** shows a rear elevation view of the fourth embodiment of the modified MOG limiting concave assembly of FIG. **29**.
(32) FIG. **31** shows a side elevation view of the first embodiment of the MOG limiting concave assembly of FIG. **23** equipped with cover plates.
(33) FIG. **32** shows a side elevation view of the second embodiment of the MOG limiting concave assembly of FIG. **25** equipped with cover plates.
(34) FIG. **33** shows a side elevation view of the three embodiment of the MOG limiting concave assembly of FIG. **27** equipped with cover plates.
(35) FIG. **34** shows a side elevation view of the fourth embodiment of the MOG limiting concave assembly of FIG. **29** equipped with cover plates.
(36) FIG. **35** shows a top plan view of left-side quarter wrap (QW) cover plate assembly.
(37) FIG. **36** shows a side elevation view of left-side quarter wrap (QW) cover plate assembly of FIG. **35**.
(38) FIG. **37** shows a top plan view of right-side quarter wrap (QW) cover plate assembly.
(39) FIG. **38** shows a side elevation view of right-side quarter wrap (QW) cover plate assembly of FIG. **37**.
(40) FIG. **39** shows a side elevation view of a quarter wrap (QW) MOG limiter assembly included

in the MOG limiting concave assembly of FIG. **23**.
(41) FIG. **40** shows a top plan view of left-side quarter wrap (QW) MOG limiters in the quarter wrap MOG limiter assembly seen in FIG. **39**.
(42) FIG. **41** shows a side elevation view of left-side quarter wrap (QW) MOG limiter assembly of FIG. **40**.
(43) FIG. **42** shows a bottom plan view of left-side quarter wrap (QW) MOG limiter assembly of FIG. **40**.
(44) FIG. **43** shows a top plan view of right-side quarter wrap (QW) MOG limiters in the quarter wrap MOG limiter assembly seen in FIG. **39**.
(45) FIG. **44** shows a side elevation view of right-side quarter wrap (QW) MOG limiter assembly of FIG. **43**.
(46) FIG. **45** shows a bottom plan view of right-side quarter wrap (QW) MOG limiter assembly of FIG. **43**.
(47) FIG. **46** shows the experimental results for the harvest of soybeans using the quarter wrap (QW) MOG limiter assemblies and/or quarter wrap (QW) cover plate assemblies described above.
(48) FIG. **47**A shows a first perspective view over-center latch assembly with a pentagonal base, according to some aspects of the present disclosure.
(49) FIG. **47**B shows a second perspective view of the over-center latch assembly of FIG. **47**A.
(50) FIG. **48**A shows a perspective view of an over-center latch assembly with rectangular base for a right-side MOG limiter or cover plate, according to some aspects of the present disclosure.
(51) FIG. **48**B shows a side elevation view of the over-center latch assembly of FIG. **48**A.
(52) FIG. **49**A shows a perspective view of an over-center latch assembly with rectangular base for a left-side MOG limiter or cover plate, according to some aspects of the present disclosure.
(53) FIG. **49**B shows a side elevation view of the over-center latch assembly of FIG. **49**A
(54) FIG. **50**A shows a first perspective view of an over-center latch with a linchpin with an adjustable loop.
(55) FIG. **50**B shows a side elevation view of the over-center latch of FIG. **50**A.
(56) FIG. **50**C shows a top plan view of the over-center latch of FIG. **50**A.
(57) FIG. **50**D shows a second side elevation view of the over-center latch of FIG. **50**A.
(58) FIG. **51**A shows a first perspective view of an attaching assembly for securing the over-center latch assemblies of FIG. **47**A to concave(s).
(59) FIG. **51**B shows a second perspective view of the attaching assembly of FIG. **51**A.
(60) FIG. **52**A shows a first perspective view of an attaching assembly for securing the over-center latch assemblies to concave(s) including the over-center latch assembly of FIG. **48**A.
(61) FIG. **52**B shows a second perspective view of the assembly of FIG. **52**A.
(62) FIG. **53**A shows a first perspective view of the over-center latch assembly of FIG. **47**A mounted to a solid, extended mounting portion that is included in a right-side quarter wrap (QW) MOG limiter assembly.
(63) FIG. **53**B shows a second perspective view thereof.
(64) FIG. **53**C shows a first perspective view of the over-center latch assembly of FIG. **47**A mounted to a solid, extended mounting portion that is included in a left-side quarter wrap (QW) MOG limiter assembly.
(65) FIG. **53**D shows a second perspective view thereof.
(66) FIG. **54**A shows a perspective view of a right-side MOG limiter including the over-center latch assembly of FIG. **48**A.
(67) FIG. **54**B shows a back elevation view of the MOG limiter of FIG. **54**A.
(68) FIG. **54**C shows an additional perspective view of the MOG limiter of FIG. **54**A.
(69) FIG. **54**D shows a bottom plan view of the MOG limiter of FIG. **54**A.
(70) FIG. **55**A shows a perspective view of a left-side MOG limiter including the over-center latch assembly of FIG. **49**A.

(71) FIG. **55**B shows a bottom view of the MOG limiter of FIG. **55**A.
(72) FIG. **55**C shows an additional perspective view of the MOG limiter of FIG. **55**A.
(73) FIG. **55**D shows a side elevation view of the MOG limiter of FIG. **55**A.
(74) FIG. **56**A shows a first perspective view of over-center latch assemblies securing two pairs of left-side and right-side MOG limiter assemblies to a concave.
(75) FIG. **56**B shows a second perspective view thereof.
(76) FIG. **57**A shows a first perspective view of a concave with over-center latch assemblies securing the MOG limiters of FIGS. **54**A and **55**A.
(77) FIG. **57**B shows an additional perspective view of the concave of FIG. **57**A.
(78) FIG. **57**C shows a bottom plan view of the concave of FIG. **57**A.
(79) FIG. **58**A shows a first perspective view of the over-center latch assembly of FIG. **47**A mounted to a triangular mounting portion that is included in a left-side quarter wrap (QW) cover plate assembly.
(80) FIG. **58**B shows a second perspective view thereof.
(81) FIG. **58**C shows a first perspective view of the over-center latch assembly of FIG. **47**A mounted to a solid, extended mounting portion that is included in a right-side quarter wrap (QW) cover plate assembly.
(82) FIG. **58**D shows a second perspective view thereof.
(83) FIG. **59**A shows a first perspective view of the over-center latch assembly of FIG. **48**A mounted to a solid, extended mounting portion that is included in a right-side quarter wrap (QW) cover plate assembly.
(84) FIG. **59**B shows a second perspective view thereof.
(85) FIG. **59**C shows a side elevation view thereof.
(86) FIG. **59**D shows a bottom plan view thereof.
(87) FIG. **60**A shows a first perspective view of the over-center latch assembly of FIG. **49**A mounted to a solid, extended mounting portion that is included in a left-side quarter wrap (QW) cover plate assembly.
(88) FIG. **60**B shows a second perspective view thereof.
(89) FIG. **60**C shows a side elevation view thereof.
(90) FIG. **60**D shows a bottom elevation view thereof.
(91) FIG. **61** shows a first perspective view of over-center latch assemblies securing two pairs of left-side and right-side cover plate assemblies to a concave.
(92) FIG. **62** shows a second perspective view thereof.
(93) FIG. **63** shows a first perspective view of over-center latch assemblies securing two pairs of the cover plate assemblies shown in FIG. **59**A and two pairs of the cover plate assemblies shown in FIG. **60**A to a concave.
(94) FIG. **64** shows a second perspective view thereof.
(95) FIG. **65** shows a bottom plan view thereof.
(96) FIG. **66** shows a locking nut with a nylon friction ring incorporated therein.
(97) An artisan of ordinary skill in the art need not view, within isolated figure(s), the near infinite distinct combinations of features described in the following detailed description to facilitate an understanding of the present disclosure.

DETAILED DESCRIPTION

(98) The present disclosure is not to be limited to that described herein. Mechanical, electrical, chemical, procedural, and/or other changes can be made without departing from the spirit and scope of the present disclosure. No features shown or described are essential to permit basic operation of the present disclosure unless otherwise indicated.
(99) FIG. **1** shows a typical combine harvester **50** which may utilize the embodiments described herein. The combine harvester **50** typically includes a header **52** to interact with crops, a conveyer **54** to take the crops that have been cut by the header **52** into the combine **50**, a threshing chamber

56 to receive the cut crops from the conveyer, the threshing chamber **56** containing a rotor surrounded by a cage of concaves to separate grain from the crop, a cleaning system **58** located below the threshing chamber to catch separated grain and any MOG (material other than grain) on sieves which filter grain through the sieves and release MOG out a back of the combine, and a cab **60** for a user to sit in and operate the combine. As will be described herein, systems, kits, and devices of the present disclosure relate to components within the threshing chamber **56** to improve the combine harvester's **50** performance.
(100) Rotor combines utilize an internal chamber wherein the harvested plant material including the grain is passed from the head of the combine into the threshing chamber or rotor chamber. This rotor chamber processes the material by a threshing operation and a concurrent separation operation. A series of curved apparatus are positioned at the bottom of the rotor chamber whereat grain is allowed to drop out of the chamber into the cleaning system which operates to clean the grain then convey and store the grain in the grain bin or tank in the combine. As an example of this set up, the John Deere single rotary combines have seven curved apparatus running from the front of the chamber rearwardly and positioned side-by-side in positions #1-#7. Positions #1-#3 (positioned closest to the header, see, e.g., FIG. **22**) are customarily referred to as the threshing section and are occupied by threshing concaves, and positions #4-#7 (positioned further away from the header along the cage surrounding the rotor) are customarily referred to as the separation section and are occupied by separation grates.
(101) The symptoms indicative of inefficiency in the harvest of small grains such as soybeans include the presence of unthreshed pods in the grain tank or bin of the combine, excessive return tailings, MOG in the grain tank, split beans, and/or sieve loss. A primary contributor to these symptoms involves the combine which historically has utilized a single concave design in the threshing of multiple grains, larger and smaller in size.
(102) The combine concaves are a key part of a combine harvester, with two vital roles in the harvesting process: freeing seeds from plants (threshing), and moving them away from the chaff (winnowing). The quality of the harvested crop depends significantly on the quality and configuration of combine concaves in use which form the cage surrounding the rotor within the threshing chamber.
(103) As shown in FIGS. **2-5**, and FIGS. **56**A-**57**C, **61-65**, concaves **101**, **500**, **600**, **700**, **800** are elongated, heavy, and curvilinear. In a combine, several concaves **101**, **500**, **600**, **700**, **800** work in tandem to thresh and separate grain from MOG and are positioned in series. The concaves **101**, **500**, **600**, **700**, **800** represent a substantial expense if differing concave designs are used by the farmer. Further, concaves **101**, **500**, **600**, **700**, **800** are difficult and time-consuming to change out in a combine. At harvest, time is critical to take-in the crop at the optimal time as conditions for harvest change with the weather. Thus, the dilemma presented to the farmer is which concave design to use within the combine and whether to change-out the concaves or live with what is in place when the combine is originally set-up for harvest.
(104) For example, concaves **101**, **500**, **600**, **700**, **800** that are used in the harvest of smaller grains can include narrower gaps and/or openings to limit the amount of MOG that drops out of the chamber and into the cleaning system. Concaves **101**, **500**, **600**, **700**, **800** that are intended for the harvest of larger grains should include gaps that still allow the larger grains to drop out of the chamber and into the cleaning system.
(105) The concaves **101**, **500**, **600**, **700**, **800** are generally constructed from stainless steel or plastic. The materials that are used to construct the concaves **101**, **500**, **600**, **700**, **800** should be high-quality durable materials that stand up to wear and tear over repeated use without losing their threshing efficiency.
(106) The concaves **101**, **500**, **600**, **700**, **800** are also preferably kept in good condition so that they can effectively separate grain from chaff without becoming clogged or damaged by stray stones or other foreign objects in the field. Maintenance of the concaves **101**, **500**, **600**, **700**, **800** can be a

key part of achieving optimal performance with the concaves **101**, **500**, **600**, **700**, **800**. The concaves **101**, **500**, **600**, **700**, **800** must not only be kept free from debris, but also the farmer should periodically evaluate whether the openings in the concaves **101**, **500**, **600**, **700**, **800** are an appropriate size for the crop to be harvested.

(107) However, because of the aforementioned hardship(s) associated with changing out the concaves **101**, **500**, **600**, **700**, **800**, it is highly desired to be able to use concaves **101**, **500**, **600**, **700**, **800** that are suitable for both larger and smaller grains. One such solution is to always use concaves **101**, **500**, **600**, **700**, **800** that are suitable for larger grains, and to attach removable inserts **105**, **1100**, **1200** and/or cover plates **300**, **400**, **900**, **1000** which narrow the gaps and/or openings when harvesting smaller grains. The removable inserts **105**, **1100**, **1200** and/or cover plates **300**, **400**, **900**, **1000** could, for example, attach to the concaves **101**, **500**, **600**, **700**, **800** at upper and lower connection points **102**, **102**A shown in FIG. **2**.

(108) The cover plates **300**, **400**, **900**, **1000** retain plant material to fill the large openings and create a mattress-type surface. The matte of plant material provides a gentle & natural rubbing action against each other as they travel over the horizontal grain separation bars while retaining the material in chamber for enough time to complete even the tough or green pods. The cover plates **300**, **400**, **900**, **1000** also allow the operator to run the bottom sieve further open, pushing more air through the top sieve, reducing pod material in the tank. The use of natural plant materials reduces the chances of the rotor splitting soybeans over the longer duration in the chamber.

(109) Use of the cover plates **300**, **400**, **900**, **1000** allows for a more complete thresh, reduces tailings, and creates a cleaner grain sample.

(110) A condition addressed by embodiments of the present disclosure involve the harvest of high moisture grain. In such conditions, the grain is more likely to clump or bridge the gap between concave crossbars. In such situations, MOG limiters can quickly go from helping eliminate MOG in harvested material to plugging of said gaps.

(111) In breakdown of the threshing section of the rotor chamber of a combine, experimentation and study with the harvest of soybeans has shown that historically pods fall through the concave in the forward or #1 position due in substantial part to the failure to retain the pods within the rotor chamber to allow the pod-on-pod abrading action created by the revolving rotor and attached rasp bars which functions to open the pods and separate the beans therein. Excessive amounts of MOG historically are the function of MOG not being retained within the rotor chamber and falling through the concaves at positions #2 and #3. If OEM filler inserts are used in positions #2 and/or #3, the resultant problem is often loss of harvest yield due to the retention of the beans within the rotor chamber resulting in abrasive fracturing or crushing or their loss out the back end of the combine with the MOG.

(112) The present disclosure is designed to address the issue of retaining MOG within the rotor chamber by metering the removal of grain from the rotor chamber. The MOG limiters thereby allow for the timely and efficient removal of the beans or other small grains, including small kernel corn, from the threshing portion of the rotor chamber of the combine. The more quickly the separated grain is removed from the rotor chamber, the lesser the process load is placed on the threshing and separation functions which increases the overall efficiency of the harvest process while maintaining heightened integrity of the grain.

(113) The present disclosure allows the farmer to balance the load of the harvest operation in a variety of different ways. For example, several embodiments disclosed herein allow for more control over the volume of MOG dropping out of the rotor chamber to the sieve area.

(114) In another respect, the present disclosure allows for a balance of grain throughput from rotor chamber to the sieve area. This balance relates to the constant volume flow of grain deposited through the concaves on left side versus right side. Historically, experimentation and study show that rotary combines deposit grain out of the rotor chamber heavily on the right side in what may be referred to as the 3-6 o'clock position when viewed from a position standing behind the combine.

As the grain is transferred back to the cleaning area wherein the sieve is positioned, the right side of the sieve receives the majority of the material that exited the rotor chamber for cleaning while the left side of the sieve receives substantially lesser material for cleaning. Thus, the right side of the sieve is overloaded which negatively impacts its efficiency of properly separating grain from the remaining MOG and allows a greater amount of grain to be lost out of the back end of the combine. With the present disclosure, the farmer can fine tune the combine to balance out the loading of the back end to improve grain cleaning and retention for deposit in the grain tank.
(115) Referring now to the assembly **100** shown throughout FIGS. **2-9**, the MOG limiter **105** is comprised of a series of equally spaced gap bars or inserts **106**, often in a rectangular configuration, positioned on a face side of the MOG limiter **105**.
(116) The series of equally spaced gap bars or inserts **106** are retained by dual, parallel rail supports **107** having a curvilinear geometry. The radius of curvature of the face side of the rail supports **107** approximates the radius of curvature of an underside of the concave **101**. The inserts **106** are vertical to the rail supports **107** and are positioned to fit up and into the gap between the concave crossbars **110** to create controlled openings.
(117) According to a preferred embodiment, the thickness of the rails is approximately 1 inch. According to yet another embodiment in need of an increase in concave open area (COA), the thickness of the rails is approximately ¼ inch where the distance in the outer surfaces of the rail supports **107** remains the same.
(118) In one embodiment, the face side of the MOG limiter is attached to the underside of the concave **101** utilizing on one end a hinge point at the end of each rail support **107** which is insertably retained in a slotted opening of a crossbar member positioned and affixed to the bottom side of the concave **101**. On the other end, a mounting plate **108** is used through which a bolt positioned on the concave **101** protrudes with retention accomplished by use of a washer and nut assembly.
(119) In one embodiment an over-center latch assembly is utilized in cooperation with the mounting plate **108**.
(120) In one embodiment the attachment means is adjustable so as to slidably move the inserts within the gaps between the concave crossbars so as to further control the opening remaining within said gaps.
(121) In the embodiments rounded notches **109** can be made on the face side of each rail support **107** which conform to the rounded geometry of the crossbars **110** on the concave **101**, and function to position the notched surface firmly against the underside of the concave **101** round crossbar **110** depending on where the rounded notches **109** are positioned.
(122) In the embodiment as shown in FIGS. **2-5**, the MOG limiters **105** are of a length which approximates one-half of the curvilinear length of the concave **101**. A plurality of bolt, washer and nut assemblies protrude downwardly from the underside of the concave **101** with a single assembly positioned in the middle of each side as shown on FIGS. **2-3** which in cooperation with the mounting plate **108** on the MOG limiters **105** form an upper concave connection **102** and a lower concave connection **102**A in each half of the concave **101** underside. The MOG limiter **105** may be positioned between a side rail **103**A and a center rail **103**B. In the cross frame **103**C are positioned openings such as insert slots **104** to receive the hinge point **111** located at the end of each rail support **107**. The hinge points **111** are slidably inserted into the insert slots **104** (as shown in FIG. **5**) and then rotated upwardly to be positioned adjacent to and in conformity with the curvature of the underside of the concave **101** with the connection assemblies **102** & **102**A holding the MOG limiter **105** in place. As shown in FIG. **4**, bolts **112** can be used to secure the **108** mounting plate to the MOG limiter **105**. The mounting plate **108** can directly attached to the axial bar **106**, one or more of the rail supports **107**, and/or additional supports that are added to the system to make for more robust securement.
(123) FIG. **5** emphasizes a view of another example of mounting hardware that can be used to

attach the MOG limiter **105** to the concave **101** as depicted in FIGS. **2-6**. For example, a mounting plate **200** is shown receiving a threaded post **201** that allows for a J-hook **202** to attach thereto. The J-hook includes a straight end (the top of the "J") and a hooked end (the bottom of the "J"). The J-hook **202** is threaded at the straight end. In the embodiment shown, the threads are male threaded configured to receive a female threaded hexagonal sleeve **203**, which is either directly and/or indirectly attached to the lever **204**.

(124) The lever **204** comprises a beam or rigid rod pivoted at a fixed hinge, or fulcrum. The rigid body of the lever **204** is capable of rotating about said fulcrum, which in this case can be the hexagonal sleeve **203**, an indirect attachment that connects to said sleeve **203**, and/or any other suitable component which allows pivotal movement such that the J-hook **202** can be move in a rotational direction. After being moved in a rotational direction by the lever **204**, the J-hook **202** can latch onto the threaded post **201**, and in the embodiment shown in FIG. **7**, indirectly to the mounting plate **200**. On the basis of the locations of the sleeve **203**, load from the MOG limiter **105**, and effort required to be input by the user, the lever **204** is designed to amplify the input force from a user to provide a greater output force. In other words, the user provides leverage and gains a mechanical advantage in that the ratio of the output force to the input force is positive.

(125) The embodiment disclosed in FIGS. **6-7** includes a face of the rail supports **107** that is smooth and allows the face to slide against the underside of the crossbars **10** of the concave **101** so that an adjustment mechanism at one end of the MOG limiter may operate to adjust and slidably position the inserts **106** within the gap to allow for greater control of the size of the gap and, thus, greater control by the farmer as to the operations of the threshing operation in the combine based upon weather, field conditions, crop conditions and grain type.

(126) The MOG limiter **105** shown in FIG. **8** positions the notches **109** adjacent to the inserts **106** which operates to position the inserts **106** adjacent to the crossbars **110** of the concave **101** creating a more restricted area on the side nearest to the crossbar **110** and a greater opening in the remaining unrestricted gap area.

(127) The MOG limiter **105** shown in FIG. **9** involves positioning these notches **109** approximately equal distance between the inserts **106** of the MOG limiter which facilitates the positioning of the inserts **106** in the middle of the gaps of the concave **101** thereby restricting the opening allowing for small grains to drop through the concave **101** on either side of the insert while preventing MOG from passing through.

(128) The MOG limiter **105** as shown in FIGS. **6-9** are readily installed by the farmer or technician without requiring the assistance of a second person and are easily and quickly removed once installed. This functions to enhance the flexibility provided to the farmer who may make adjustments as to how many and where MOG limiters **105** are positioned within the threshing section of the rotor chamber.

(129) In one embodiment a known latch assembly is utilized as the upper concave connection and lower concave connection. The latch assembly may also employ a spring tensioning handle which upon closing draws the MOG limiter up into position on the underside of the concave **101**.

(130) In one embodiment, the MOG limiter **105** is designed for use on a ⅝ inch round bar concave **101** having a ¾ inch gap between the round crossbars **110**.

(131) In one embodiment, the MOG limiter **105** utilizes a clip apparatus which inserts over the concave crossbar **110** near midpoint of the concave **101** to which it is affixed.

(132) In one embodiment, the MOG limiter **105** utilizes a pivot joint to rotatably affix the end of the MOG limiter at or near the middle of the concave **101**.

(133) In one embodiment, the MOG limiter **105** utilizes an insertable T-slot pivot assembly or other known standard pivot configurations to affix and retain the one end of the MOG limiter at or near the center point of the concave **101**.

(134) In one embodiment, the MOG limiter utilizes at least one clip mechanism to fasten the end of the MOG limiter to the end of the concave **101** and to retain the MOG limiter against the curvature

of the concave **101**.
(135) In one embodiment, the clip assembly of the MOG limiter **105** is adjustable thereby pushing or pulling the MOG limiter **105** in a lateral direction as it is positioned on the concave **101**.
(136) In one embodiment the MOG limiter is designed for use with a ⅝ inch round bar concave **101** having a ⅞ inch gap between the round crossbars **110**.
(137) In one embodiment, the MOG limiter **105** utilizes at least one threaded locking mechanism to fasten the end of the MOG limiter to the end of the concave **101** and to retain the MOG limiter against the curvature of the concave **101**.
(138) In one embodiment, the threaded locking mechanism of the MOG limiter may be pulled or pushed by turning a retention housing or one or more retention nuts, thereby moving the MOG limiter laterally as it is positioned on the concave **101**.
(139) In one embodiment, the MOG limiter **105** is mechanically moved laterally thereby positioning the as it utilizes at least one threaded locking mechanism to fasten the end of the MOG limiter **105** to the end of the concave **101** and to retain the MOG limiter against the curvature of the concave **101**.
(140) Multiple other attachment mechanisms known in the art may be used to attach and hold the MOG limiter **105** in a stationary position adjacent to the concave **101** and which may allow for the adjustment of the position of the inserts **106** of the MOG limiter within the gaps existing between the crossbars **110** of the concaves residing within the combine.
(141) Other embodiments of the MOG limiter **105** may employ inserts **106** of greater or lesser width so as to increase or decrease the openings remaining with the MOG limiters such that adjustments may be made by the operator of the combine. For example, in the harvest of sunflower seeds, the operator may set the MOG limiter to provide for a larger gap. Alternatively, in the harvest of alfalfa the opening selected may be set at or nearest to its narrowest. This may be accomplished by positioning the inserts equal distance from the crossbars to create two equal or nearly equal openings within each gap.
(142) In a concave **101** having a center support element on the underside as shown in FIG. **2** and utilizing the MOG limiter **105** as depicted in one of FIGS. **6-9**, there are four quadrants existing on the underside of the concave **101**. The farmer may install MOG limiters in all four quadrants of the concaves **101** at positions #2 and #3 to enhance the retention of MOG within the rotor chamber of the combine.
(143) A method of use of the MOG limiters **105** in the 3-6 o'clock position on the concaves **101** placed in positions #2 &/or #3 to control the flow and distribution of grain to the sieve below so as to maximize the cleaning of the grain with the minimum of grain loss.
(144) The assembly **100** can also include the use of one or more covers **300**, **400**, as is shown throughout FIGS. **10-13**. Specifically, an example configuration for a left-side cover is shown throughout FIGS. **10-11** and an example configuration for a right-side cover is shown throughout FIGS. **12-13**.
(145) The left-side cover **300** can include a curvilinear plate **301**. The curvilinear plate **301** comprises a flat, opaque, and rigid material that can be used to near completely prevent the flow of MOG from dropping out of the chamber and into the cleaning system. The curvilinear plate **301** is positioned between an arcuate side rails **107** and middle support rail of a concave **101**. The cover plate **300** is designed to be configured in a curved supinated position against the crossbars **110** on the exterior of the concave **101**.
(146) The cover plate **300** may also include one or more arced supports **302**, **303** that provide support to the curvilinear plate **301** from behind where a load of grain and MOG is applied. The one or more arced supports **302**, **303** generally traverse the length (lateral direction) of the cover plates **300**. In some embodiments, the one or more arced supports **302**, **302** can further facilitate attachment to the arced rails **107** of the concave **101**.
(147) The latching assembly **200-204** shown and described in FIG. **5** can similarly be employed for

the latching the cover plates **300** to the concave **101**. The J-hook **202** in particular can maintain the lateral position of the cover plate so that curvilinear plate **201** stays adjacent the axial bars **110**, thereby stopping the flow of MOG therethrough. The slot **304** is the area of the curvilinear plate **301** that allows for the threaded post **201** to pass through. The slot is preferably centrally located within a mounting plate **305** included on the cover plate **300**.

(148) Other supports that can also be included are bridge supports **306**. The bridge supports **306** support the curvilinear plate **301** from behind where a load of grain and MOG is applied. These bridge supports generally span the horizontal length of the cover plate **300** from the one or more arced supports **302**, **303** to the other. In some embodiments, several are employed within a single cover plate **200** one or more arced supports **302**, **300** and they are spaced throughout the lateral distance of the cover plate **300** (see, e.g., FIGS. **10-11**).

(149) Likewise, the right-side cover **400** can include a curvilinear plate **401**. The curvilinear plate **401** comprises a flat, opaque, and rigid material that can be used to near completely prevent the flow of MOG from dropping out of the chamber and into the cleaning system. The curvilinear plate **401** is positioned between an arcuate side rails **107** and middle support rail of a concave **101**. The cover plate **400** is designed to be configured in a curved supinated position against the crossbars **110** on the exterior of the concave **101**.

(150) The cover plate **400** may also include one or more arced supports **402**, **403** that provide support to the curvilinear plate **401** from behind where a load of grain and MOG is applied. The one or more arced supports **402**, **403** generally traverse the length (lateral direction) of the cover plates **400**. In some embodiments, the one or more arced supports **402**, **402** can further facilitate attachment to the arced rails **107** of the concave **101**.

(151) The latching assembly **200-204** shown and described in FIG. **5** can similarly be employed for the latching the cover plates **400** to the concave **101**. The J-hook **202** in particular can maintain the lateral position of the cover plate so that curvilinear plate **201** stays adjacent the axial bars **110**, thereby stopping the flow of MOG therethrough. The slot **304** is the area of the curvilinear plate **401** that allows for the threaded post **201** to pass through. The slot is preferably centrally located within a mounting plate **405** included on the cover plate **400**.

(152) Other supports that can also be included are bridge supports **406**. The bridge supports **406** support the curvilinear plate **401** from behind where a load of grain and MOG is applied. These bridge supports generally span the horizontal length of the cover plate **400** from the one or more arced supports **402**, **403** to the other. In some embodiments, several are employed within a single cover plate **200** one or more arced supports **402**, **400** and they are spaced throughout the lateral distance of the cover plate **400** (see, e.g., FIGS. **12-13**).

(153) In some embodiments, cover plates **300**, **400** are installed in the concave **101** located in position #1.

(154) It is thus to be appreciated that depending on the application, various combinations of MOG limiters **105**, left-side cover plates **300**, and right-side cover plates **400** can be employed to more uniformly distribute grain and MOG in the threshing and separation chambers of the combine.

(155) For example, in the embodiment shown throughout FIGS. **14-17**, a single MOG limiter is employed at the top left quarter of the concave **101**, a single left-side cover plate **300** is employed at the lower left quarter of the concave **101**, and two right side cove plates **400** are employed at the upper right quarter and the lower right quarter of the concaves.

(156) For example, in the embodiment shown throughout FIGS. **18-21**, a single MOG limiter is employed at the top right quarter of the concave **101**, a single right-side cover plate **400** is employed at the lower right quarter of the concave **101**, and two left side cove plates **300** are employed at the upper right quarter and the lower right quarter of the concaves.

(157) It should further be appreciated combines including one or more assemblies (**100**) can include assemblies of different configurations to facilitate a more uniform grain and MOG distribution in the threshing and separation chambers of the combine. Thus, the configuration

shown in FIGS. **14-17** could be used in position #1, while the configuration shown in FIGS. **18-21** could be used in position #2. An assembly having only MOG limiters could be used in position #3.
(158) The left-side and right side covers **300**, **400** are configured to compliment one another such that ends of the arced rails **302**, **303**, **402**, **403** and/or separate pins which extend therefrom do not interfere with one another and/or interlock as they pass through the insert slots **104** within the cross frame **103**C. In some embodiments, it is preferred that the left-side and right-side cover plates **300**, **400** are not identical and/or mirror images of one another to help establish the aforementioned interlock.
(159) It is to be appreciated that, in some embodiments, the left-side and right-side cover plates **300**, **400** can be identically configured such that by simply rotating one 180°, a left-side cover plate **300** becomes a right-side cover plate **400** and vice-versa.
(160) In some embodiments, it should be appreciated that the concave can be divided into six, eight, or ten equal sections instead of the four quarter sections that are shown throughout FIGS. **14-21**. One example combination is suggested in FIG. **22**, in which the combination portrayed in FIG. **14** is shown in the #1 position of FIG. **22**. Further, for orientation purposes, a directional continuum arrow generally presents how the concaves would be placed within a combine (such as that shown in FIG. **1**). As shown, moving in a direction of lower concave positions indicates moving closer to a header of the combine (e.g., moving from position #2 to position #1), whereas moving in a direction of higher concave positions (e.g., positions #2 to position #3, or position #6 to position #7 not shown) indicates moving closer to a back of the combine.
(161) A first embodiment **500** of a modified concave is shown in FIGS. **23-24**. The first embodiment **500** includes an enhanced side flow rail **501**, a C-clamp end **502**, a first-end round bar drive **503**, a second end plate **504**, a quarter wrap center catch **505**, an enhanced flow center rail **506**, 0.625 inch diameter round bars **507**, a left over-center catch **508**, and a right over-center catch **509**.
(162) The side flow rails **501** are the curvilinear component that serve as the outer frame of the concave. At the C-clamp end **502** of the side flow rail **501**, there exists a C-clamp that extends outwardly from a first-end round bar drive **503**. At the second end there exists a second end plate **504**. The C-clamp end **502** and the second end plate **504** are spaced from one another and extend generally away from one another. The C-clamp end **502** and the second end plate **504** extend transversely between and are rigidly joined to corresponding ends of the curved by way of the side flow rails **501**. The C-clamp end **502** is curved in configuration for supporting the side flow rails **501**, but there could exist some straight configurations for supporting the side flow rails **501** as part of the casing (not shown) below the rotor (not shown) of the combine.
(163) The first embodiment **500** includes major central supports in the quarter wrap center catch **505** and the enhanced flow center rail **506**. The quarter wrap center catch **505** is roughly equidistant between the C-clamp end **502** and the first-end round bar drive **503**. The quarter wrap center catch **505** in combination with the left-over-center catch **508** and right-over-center catch **509** facilitates secure attachment to cover plates **900**, **1000** and MOG limiters **1100**, **1200**.
(164) The enhanced flow center rail **506** runs parallel to the side flow rails **501** and provides further support to the round bars **507**. The round bars **507** are spaced substantially equidistantly from one another so that the gaps that form therebetween comprise a substantially similar or identical dimension.
(165) A second embodiment **600** of a modified concave is shown in FIGS. **25-26**. The second embodiment **600** includes an enhanced side flow rail **601**, a C-clamp end **602**, a first-end round bar drive **603**, a second end plate **604**, a quarter wrap center catch **605**, 0.375 rub bars **606** with 0.125 inch prouds, 0.625 inch diameter round bars **607**, an enhanced flow center rail **608**, a left over-center catch **609**, and a right over-center catch **610**.
(166) The side flow rails **601** are the curvilinear component that serve as the outer frame of the concave. At the C-clamp end **602** of the side flow rail **601**, there exists a C-clamp that extends

outwardly from a first-end round bar drive **603**. At the second end there exists a second end plate **604**. The C-clamp end **602** and the second end plate **604** are spaced from one another and extend generally away from one another. The C-clamp end **602** and the second end plate **604** extend transversely between and are rigidly joined to corresponding ends of the curved by way of the side flow rails **601**. The C-clamp end **602** is curved in configuration for supporting the side flow rails **601**, but there could exist some straight configurations for supporting the side flow rails **601** as part of the casing (not shown) below the rotor (not shown) of the combine.

(167) The first embodiment **600** includes major central supports in the quarter wrap center catch **605** and the enhanced flow center rail **608**. The quarter wrap center catch **605** is roughly equidistant between the C-clamp end **602** and the first-end round bar drive **603**. The quarter wrap center catch **605** in combination with the left-over-center catch **609** and right-over-center catch **610** facilitates secure attachment to cover plates **900**, **1000** and MOG limiters **1100**, **1200**.

(168) The enhanced flow center rail **608** runs parallel to the side flow rails **601** and provides further support to the rub bars **606** and the round bars **607**. The rub bars **606** and the round bars **607** alternate. The rub bars **606** and the round bars **607** are spaced substantially equidistantly from one another so that the gaps that form therebetween comprise a substantially similar or identical dimension.

(169) A third embodiment **700** of a modified concave is shown in FIGS. **27-28**. The third embodiment **700** includes an enhanced side flow rail **701**, a C-clamp end **702**, a first-end round bar drive **703**, a second end plate **704**, a quarter wrap center catch **705**, 0.25 inch flat rub bars **706**, 0.625 inch diameter round bars **707**, an enhanced flow center rail **708**, a left over-center catch **709**, and a right over-center catch **710**.

(170) The side flow rails **701** are the curvilinear component that serve as the outer frame of the concave. At the C-clamp end **702** of the side flow rail **701**, there exists a C-clamp that extends outwardly from a first-end round bar drive **703**. At the second end there exists a second end plate **704**. The C-clamp end **702** and the second end plate **704** are spaced from one another and extend generally away from one another. The C-clamp end **702** and the second end plate **704** extend transversely between and are rigidly joined to corresponding ends of the curved by way of the side flow rails **701**. The C-clamp end **702** is curved in configuration for supporting the side flow rails **701**, but there could exist some straight configurations for supporting the side flow rails **701** as part of the casing (not shown) below the rotor (not shown) of the combine.

(171) The first embodiment **700** includes major central supports in the quarter wrap center catch **705** and the enhanced flow center rail **708**. The quarter wrap center catch **705** is roughly equidistant between the C-clamp end **702** and the first-end round bar drive **703**. The quarter wrap center catch **705** in combination with the left-over-center catch **709** and right-over-center catch **710** facilitates secure attachment to cover plates **900**, **1000** and MOG limiters **1100**, **1200**.

(172) The enhanced flow center rail **708** runs parallel to the side flow rails **701** and provides further support to the rub bars **706** and the round bars **707**. The rub bars **706** and the round bars **707** alternate. The rub bars **706** are included in triplets and the remaining bars are round bars **707**. The rub bars **706** and the round bars **707** are spaced substantially equidistantly from one another so that the gaps that form therebetween comprise a substantially similar or identical dimension.

(173) A fourth embodiment **800** of a modified concave is shown in FIGS. **29-30**. The fourth embodiment includes an enhanced side flow rail **801**, a C-clamp end **802**, a first-end round bar drive **803**, a second end plate **804**, a quarter wrap center catch **805**, 0.375 inch flat rub bars **806**, 0.625 inch diameter round bars **807**, an enhanced flow center rail **808**, a left over-center catch **809**, and a right over-center catch **810**.

(174) The side flow rails **801** are the curvilinear component that serve as the outer frame of the concave. At the C-clamp end **802** of the side flow rail **801**, there exists a C-clamp that extends outwardly from a first-end round bar drive **803**. At the second end there exists a second end plate **804**. The C-clamp end **802** and the second end plate **804** are spaced from one another and extend

generally away from one another. The C-clamp end **802** and the second end plate **804** extend transversely between and are rigidly joined to corresponding ends of the curved by way of the side flow rails **801**. The C-clamp end **802** is curved in configuration for supporting the side flow rails **801**, but there could exist some straight configurations for supporting the side flow rails **801** as part of the casing (not shown) below the rotor (not shown) of the combine.
(175) The first embodiment **800** includes major central supports in the quarter wrap center catch **805** and the enhanced flow center rail **808**. The quarter wrap center catch **805** is roughly equidistant between the C-clamp end **802** and the first-end round bar drive **803**. The quarter wrap center catch **805** in combination with the left-over-center catch **809** and right-over-center catch **810** facilitates secure attachment to cover plates **900**, **1000** and MOG limiters **1100**, **1200**.
(176) The enhanced flow center rail **808** runs parallel to the side flow rails **801** and provides further support to the rub bars **806** and the round bars **807**. The rub bars **806** and the round bars **807** alternate. The rub bars **806** are included every third bar and the remaining bars are round bars **807**. The rub bars **806** and the round bars **807** are spaced substantially equidistantly from one another so that the gaps that form therebetween comprise a substantially similar or identical dimension.
(177) As shown in FIG. **31**, the first embodiment **500** of the MOG limiting concave assembly of FIG. **23** can be equipped with cover plates **900/1000**.
(178) As shown in FIG. **32**, the second embodiment **600** of the MOG limiting concave assembly of FIG. **25** can be equipped with cover plates **900/1000**.
(179) As shown in FIG. **33**, the third embodiment **700** of the MOG limiting concave assembly of FIG. **27** can be equipped with cover plates **900/1000**.
(180) As shown in FIG. **34**, the fourth embodiment **800** of the MOG limiting concave assembly of FIG. **29** can be equipped with cover plate assemblies **900/1000**.
(181) FIGS. **35-36** show a left-side quarter wrap (QW) cover plate assembly **900**. The left-side quarter wrap (QW) cover plate assembly **900** includes a left-side quarter wrap cover plate **901**, a left-side quarter wrap cover plate support **902**, a left-side cover plate over-center handle anchor plate **903**, a left-side quarter wrap cover plate rail **904**, a J-bolt **905**, a 0.3125 inch 18-thread long-steel coupling nut **906**, a 0.25 inch 18-thread long hex head screw **907**, a 0.25 inch 18-thread nylon insert lock nut **908**, a 0.25 inch steel washer **909**, and a CMF over-center handle **910**.
(182) FIGS. **37-38** show a right-side quarter wrap (QW) cover plate assembly **1000**. The right-side quarter wrap (QW) cover plate assembly **1000** includes a right-side quarter wrap cover plate **1001**, a right-side quarter wrap cover plate support **1002**, a right-side cover plate over-center handle anchor plate **1003**, a right-side quarter wrap cover plate rail **1004**, a J-bolt **1005**, a 0.3125 inch 18-thread long-steel coupling nut **1006**, a 0.25 inch 18-thread long hex head screw **1007**, a 0.25 inch 18-thread nylon insert lock nut **1008**, a 0.25 inch steel washer **1009**, and a CMF over-center handle **1010**.
(183) The left-side and right-side quarter wrap (QW) cover plate assemblies **900**, **1000** retain plant material to fill the large openings and create a mattress-type surface. The matte of plant material provides a gentle & natural rubbing action against each other as they travel over the horizontal grain separation bars while retaining the material in chamber for enough time to complete even the tough or green pods. The left-side and right-side quarter wrap (QW) cover plate assemblies **900**, **1000** also allow the operator to run the bottom sieve further open, pushing more air through the top sieve, reducing pod material in the tank. The use of natural plant materials reduces the chances of the rotor splitting soybeans over the longer duration in the chamber.
(184) Use of the cover plates **900**, **1000** allows for a more complete thresh, reduces tailings, and creates a cleaner grain sample.
(185) As shown in FIG. **39**, the first embodiment **500** of the MOG limiting concave assembly of FIG. **23** can be equipped with MOG limiter assemblies **1100/1200**.
(186) FIGS. **40-42** show a left-side quarter wrap (QW) MOG limiter assembly **1100**. The left-side quarter wrap (QW) MOG limiter assembly **1100** includes a left-side quarter wrap MOG limiting

rail **1101**, a left-side MOG limiting rub bar **1102**, a left-side ML over-center handle anchor plate **1103**, a J bolt **1104**, a 0.3125 inch 18-thread long-steel coupling nut **1105**, a 0.25 inch 18-thread long hex head screw **1106**, a 0.25 inch 18-thread nylon insert lock nut **1107**, a 0.25 inch steel washer **1108**, and a CMF over-center handle **1109**.

(187) FIGS. **43-45** show a right-side quarter wrap (QW) MOG limiter assembly **1200**. The right-side quarter wrap (QW) MOG limiter assembly **1200** includes a right-side quarter wrap MOG limiting rail **1201**, a right-side MOG limiting rub bar **1202**, a right-side ML over-center handle anchor plate **1203**, a J bolt **1204**, a 0.3125 inch 18-thread long-steel coupling nut **1205**, a 0.25 inch 18-thread long hex head screw **1206**, a 0.25 inch 18-thread nylon insert lock nut **1207**, a 0.25 inch steel washer **1208**, a CMF over-center handle **1209**.

(188) The QW wrap MOG limiter assemblies **1100**, **1200** can reduce the size of the openings between bars between 30% and 50%, more preferably between 35% and 45%, and most preferably, by approximately 42%. This limits the flow of excessive MOG leaving the rotor chamber and falling into the auger bed. Because the auger bed now contains a higher percentage of grain and a lower amount of MOG, less air will be required to separate the MOG from the grain on the top sieve. Consistent results in all soybean harvesting moisture and field conditions with lower sieve loss. Ultimately, fewer soybeans are walked out the back of the combine as sieve loss.

(189) The QW wrap MOG limiter assemblies **1100**, **1200** match a custom plate built into the mid-point of the concave's arc which allows for a one man install. The QW wrap MOG limiter assemblies **1100**, **1200** are easily installed midday to harvest soybeans and can later be removed quickly for a switch back to corn. The modified concaves **500**, **600**, **700**, **800** address the needs of busy farmers who need proof of performance in hundreds of hours of lab and field testing. After installing the QW wrap MOG limiter assemblies **1100**, **1200**, a cleaner sample and lower sieve loss is achieved when harvesting soybeans and small grains.

(190) FIGS. **47**A-**49**B show variations of an over-center latch assembly **1300**, **1400** which may each be categorized into right over-center latch assemblies **1300**R, **1400**R and left over-center latch assemblies **1300**L, **1400**L. The over-center latch assembly **1300**, **1400** includes a base plate **1301**, **1401**, a handle **1302**, **1402**, a lock hole **1303**, **1403**, a lever **1304**, **1404**, a loop **1305**, **1405**, a center pin **1306**, **1406**, a center pin **1307**, **1407**, a slot **1308**, **1408**, holes **1309**, **1409**, nylon threaded locknuts **1310**, **1410**, a subbase **1311**, **1411**, and a locking mechanism **1412**.

(191) To assemble the over-center latch assembly **1300**, **1400**, the base plate **1301**, **1401** acts as a base to which the subbase **1311**, **1411** may attach to via the holes **1309**, **1409**. The holes **1309**, **1409** may be threaded so as to receive screws, threaded rods, etc. to insert into the subbase **1311**, **1411**. Additionally, the base plate **1301**, **1401** may be attached to the subbase **1311**, **1411** via weld such as via plug weld through the holes **1309**, **1409** to fasten to the subbase **1311**, **1411**. Alternatively, the subbase **1311**, **1411** may be entirely removed such that components of the over-center latch assembly **1300**, **1400** extend from the base plate **1301**, **1401** itself.

(192) The subbase **1311**, **1411** includes cars **1307**, **1407** that extend therefrom for attaching to the handle **1302**, **1402**. Additionally, the subbase **1311**, **1411** includes the lock hole **1303**, **1403** extending away from the subbase **1311**, **1411** in generally a same direction as the cars **1307**, **1407**. The cars **1307**, **1407** and the locking hole **1303**, **1403** are configured to extend from the subbase **1311**, **1411** in such a way that the handle **1302**, **1402** may pivotably attach at two or more of the cars **1307**, **1407** acting as a fulcrum for the handle **1302**, **1402** and wherein the locking hole **1303**, **1403** extends past the handle when a distal end of the handle is positioned nearest the base plate **1301**, **1401** so as to be capable of receiving the locking mechanism **1412** through the locking hole **1303**, **1403** in such a way that it retains the distal end of the handle **1302**, **1402** adjacent the base plate **1301**, **1401** (note that the over-latch assembly **1300** may also include the locking mechanism **1412** even though one is not shown in the figures). As mentioned above, the subbase **1311**, **1411** may be removed in its entirety, in which case the cars **1307**, **1407** and the locking hole **1303**, **1403** would extend directly from the base plate **1301**, **1401**.

(193) As mentioned above, the handle **1302**, **1402** connects at its base to the cars **1307**, **1407** and as such forms a pivotable connection at the cars **1307**, **1407**. A connection joint can be seen in FIGS. **48**A-**50**C in which a fastener extends through both the cars **1407** and the handle **1402** which can especially be seen in FIG. **50**A. This connection joint is not shown in FIGS. **47**A-**47**B with the over-center latch assembly **1300** such that apertures within the cars **1307** and the handle **1302** can be seen. As understood by those of ordinary skill in the art, this does not preclude the over-latch assembly **1300** from including the connection joint shown in FIGS. **48**A-**50**C to extend through the cars **1307** and the handle **1302**. Moreover, the handle **1302**, **1402** includes an angled distal end to assist with unlocking the over-center latch assembly when grabbing the distal end of the handle **1302**, **1402**.

(194) The handle **1302**, **1402** includes a connection section for the center pin **1306**, **1406** to attach to. The connection section is positioned on the handle **1302**, **1402** at a point between the fulcrum of the handle **1302**, **1402** (connecting to the cars **1307**, **1407**) and the distal end of the handle **1302**, **1402**. The center pin **1306**, **1406** is secured at the connection section so as to be in rotational communication with the handle **1302**, **1402** to act as a fulcrum for the lever **1304**, **1404** and loop **1305**, **1405**. A base portion of the lever **1304**, **1404** may extend through the center pin **1306**, **1406** (be fixedly attached via weld, press-fit, etc., or be adjustably attached via threaded connection), wherein the lever **1304**, **1404** extends from the center pin **1306**, **1407** to include the loop **1305**, **1405** at a distal end of the lever **1304**, **1404**. As will be described herein, the loop **1305**, **1405** when assembled together with a concave will connect to the catches **1501**, **1502**, **1551**, **1552**, wherein the handle **1302**, **1402** can be lifted away from the base plate **1301**, **1401** to loosen a connection between the loop **1305**, **1405** and the catches **1501**, **1502**, **1551**, **1552**, or the handle **1302**, **1402** can be rotated towards the base plate **1301**, **1401** so as to tighten a connection and/or lock the over-center latch assembly **1300**, **1400** to the catches **1501**, **1502**, **1551**, **1552**.

(195) Should the over-center latch assembly **1300**, **1400** include a threaded connection on the lever **1304**, **1404** as shown in FIGS. **48**A-**50**C, the over-center latch assembly **1300**, **1400** may include a locking nut with a nylon insert **1410** so as to increase a holding capacity of the locking nut to stay in place and not loosen or tighten due to vibrations of the combine when in use.

(196) Further, the over-center latch assembly **1300**, **1400** may include the locking mechanism **1412** which in FIGS. **48**A-**50**C is shown as a linchpin. The linchpin is a strong holding mechanism which is given by way of example and not of limitation. Other such examples given by way of example and not of limitation include a padlock, a hairpin, a hitch pin, a spring clip, a snap ring, a wing nut assembly, etc.

(197) FIGS. **50**A-**50**D show an over-center latch **1450** without being attached to the base plate **1300**, **1400**. It is worth noting that the over-center latch **1450** uses the same numbers as those shown in FIGS. **48**A-**49**B because they are the same components shown in those figures. However, unlike the variations shown in FIGS. **48**A-**49**B, the over-center latch **1450** shown in FIGS. **50**A-**50**D does not include any numbering including an “R” or an “L” at the end thereof. Throughout FIGS. **48**A-**65**, numbering that includes the “R” indicates that the specific element being called out is side-specific to being placed on a right side of the concave when being placed on the concave. Similarly, numbering that includes the “L” indicates that the specific element being called out is side-specific to being placed on a left side of the concave when being placed on the concave. This is not limiting as to “R” must always be on a right side of the concave (or “L” always being on the left), rather this identifying of components is for simplification of understanding how the elements fit together during assembly and instead signifies that the “R” components go on a same side as one another and the “L” components go on a same side as one another, which for simplification purposes can be identified as right and left sides of the concave as described throughout this disclosure. Thus, the over-center latch **1450** does not includes any elements that are side-specific and can be used on either side on the concave depending on which base plate **1301**R, **1301**L, **1401**R, **1401**L the over-center latch **1450** is placed.

(198) Further, as shown in the FIGS. **47**A-**49**B, the over-center latch assembly **1300** does not include "R" or "L" designations while the over-center latch assembly **1400** docs. As can be seen in later figures, (e.g., FIGS. **56**A, **56**B, **57**A, and **57**B make this abundantly clear), the "R" designations include the over-center latch assemblies that go on a right side of a concave, the MOG limiter assemblies that go on the right side of the concave, and include railings along the MOG limiter that are wider apart from one another than the MOG limiter assemblies that go on a left side of the concave which include an "L" designation. Thus, while FIGS. **48**A-**48**B show the base plate **1401**R and FIGS. **49**A-**49**B show the base plate **1401**L as different, in reality the only difference is that the base plate **1401**R is wider than the base plate **1401**L so as to accommodate the MOG limiter railings to which the base plates will be attached (which is most easily seen in FIG. **57**C). As such, while FIGS. **47**A-**47**B only include the base plate **1301**, the only difference between the over-center latch assembly **1300**R and the over-center latch assembly **1300**L shown in FIGS. **56**A-**56**B is that the base plate **1301**R is wider than the base plate **1301**L because the MOG limiter **1350**L has railings that are narrower than the MOG limiter **1350**R which each attach to the cross frame of the concave. As such, it should be readily understood that a particular shape of the base plate **1301**R, **1301**L, **1401**R, **1401**L need not be specifically a pentagonal shape, a square shape, or a rectangular shape, but can include any shape that allows for connecting the over-center latch **1450** to the railings of either the MOG limiter or the cover plate and can be (given by way of example and not of limitation) a circular shape, an oval, a triangle, a rhombus, a stadium, a X-shape, etc. The variations of the base plate shown in the figures are those chosen for either of manufacturing simplicity or weight-reduction while retaining strength. Given the intended use of the base plate for the given concave, other shapes may be preferable.
(199) Furthermore, the over-center latch **1450** is shown separately from the base plates in FIGS. **50**A-**50**D to emphasize what may alternate positions on the base plates **1301**R, **1301**L, **1401**R, **1401**L. Depending on how many holes **1309**, **1409** exists on the base plates **1301**R, **1301**L, **1401**R, **1401**L the over-center latch **1450** may be positioned on either of a first end of the over-center latch assembly **1300**, **1400**, or a second end, or anywhere therebetween. Moreover as previously explained the over-center latch **1450** may have the subbase **1311**, **1411** removed entirely such that elements of the over-center latch **1450** extend directly from the base plate **1301**, **1401**. In the configurations shown in which the subbase **1311**, **1411** is included, This allows for simplicity in replacement of parts after wear or damage of the base plate **1301**, **1401**. In this regard, the over-center latch **1450** may be removed and placed on a new base plate with ease.
(200) FIGS. **51**A-**52**B show the first mounting assembly **1500** and the second mounting assembly **1550** connecting to various over-center latch assemblies **1300**R, **1300**L, **1400**R. In particular, FIGS. **51**A and **52**A each show a front view of the first mounting assembly **1500** while FIG. **52**B shows a back view of the first mounting assembly **1500**, and FIG. **51**B shows the second mounting assembly **1550**. The first mounting assembly **1500** includes a first catch **1501**, a second catch **1502**, an adjacent plate **1503**, and an end plate **1504**. The second mounting assembly **1550** includes a first catch **1551**, a second catch **1552**, and an end plate **1554**. As can be seen in later figures (e.g., FIGS. **56**A-**57**C), the first mounting assembly **1500** is attached to a right end of the concave, and the second mounting assembly **1550** is attached to a left end of the concave. The first mounting assembly **1500** and the second mounting assembly **1550** are configured to provide a structure for the over-center latch assembly **1300**, **1400** to secure itself to. Namely, extending from the end plates **1504**, **1554** are the catches **1501**, **1502**, **1551**, **1552** to which the loops **1305**, **1405** attach. Once the loop **1305**, **1405** is attached to the catch **1501**, **1502**, **1551**, **1552**, the handle **1302**, **1402** may be pressed towards the base plate **1301**, **1401** and the locking mechanism **1412** inserted into the locking hole **1303**, **1403** so as to firmly hold the assembly together.
(201) As previously stated, FIGS. **47**A-**47**B show an over-center latch assembly **1300** that can be used to help secure the left-side and right-side quarter wrap (QW) cover plate assemblies **900**, **1000** (with extended mounting portions thereon to receive the over-center latch assembly **1300**) and the

QW wrap MOG limiter assemblies **1100**, **1200** (with extended mounting portions thereon to receive the over-center latch assembly **1300**) to the concaves/concave assemblies **101**, **500**, **600**, **700**, **800**. The over-center latch assembly **1300** joins the MOG limiter and concave or the cover plate and the concave while allowing separation when unlocked. The over-center latch assembly **1300** engages another piece of hardware on another mounting surface, such as one or more of the first over-center catch **1501**, second over-center catch **1502** (see, e.g., FIGS. **51**A-**51**B), solid, extended mounting portion **1603**R (see, e.g., FIG. **53**A), solid, extended mounting portion **1703**R (see, e.g., FIG. **58**A), and solid, extended mounting portion **1703**L (see, e.g., FIG. **58**C). Depending on their design and type, catches **1501**, **1502** may also be known as strikes.
(202) The main components of the over-center latch assembly **1300** are the base plate **1301** with lever **1304** and attached loop **1305** and the handle **1302**. Tension is created once the loop **1305** is hooked onto the catches **1501**, **1502** of the mounting assembly **1500** when the lever **1304** and the handle **1302** are clamped down. Tension is released when the handle **1302** and/or the lever **1304** are pulled up into the vertical position.
(203) Because agricultural applications can result in the latches being damaged, the base plate **1301** should be constructed from a material of sufficient strength, such as mild steel, zinc coated steel and stainless steel (good for corrosive environments). The material used will affect the latches strength and durability.
(204) The over-center latch assembly **1300** includes the ability to be locked using a padlock, linchpin, or a safety catch etc. via the locking hole **1303**. This will further prevent the over-center latch assembly **1300** from accidently opening during operation.
(205) The over-center latch assembly **1300** can be adjustable in length if a threaded screw is included in the lever **1304** (see, e.g., FIGS. **50**A-**50**C). The threaded screw can be attached to the center pin **1306**. A locking nut can be ensure the threaded screw remains securely in place with respect to the center pin **1306**. For applications where the over-center latch assembly **1300** is likely to vibrate or be left in an unlocked position, a nylon friction ring can be added instead of a locking nut so that the screw loop will remain in its initial position.
(206) The loop **1305** can be an eye loop, triangle screw loop, or a hinged triangle screw loop which can be used with a variety of different catch plates. Alternatively, instead of a loop **1305**, it is to be appreciated that a T-screw, a bent T-screw, a canopy hook, clutch lever, flat hook screw, hook screw, rubber loop, or bend clamp could be utilized.
(207) The center pin **1306** include a centrally located aperture, which allows for connection to the lever **1304**. The cars **1307** extending outwardly from the base plate **1301** also include holes that allow for fasteners to be placed therethrough to form a pivoting point for the handle **1302**. The cars **1307** may in the alternative extend from a subbase **1311** (as shown) instead of directly from the base plate **1301**. In case of the cars **1307** extending from the subbase **1311**, the subbase **1311** may include apertures to align with holes **1309** to attach thereto and secure the cars **1307** to the base plate **1301**. The base plate **1301** may include a plurality of holes **1309** placed at varying distances along the base plate **1301** so as to allow for varying placement of the subbase **1311** (and thus the cars **1307**) along the base plate **1301** so as to accommodate varying distances to the catches **1501**, **1502**.
(208) The base plate **1301** can include slots **1308** to mount the base plate **1301** to a component of the left-side and right-side quarter wrap (QW) cover plate assemblies **900**, **1000** and the QW wrap MOG limiter assemblies **1100**, **1200** via a plug weld through the slot **1308**. Alternatively, the slots **1308** may be circular in shape and be threaded internally so as to allow for threaded connection to the cover plates or MOG limiters.
(209) As explained above, the base plate **1301** can include holes **1309** to attach to the subbase **1311**, but further the holes **1309** may be included at varying positions along the base plate **1301** to help balance, distribute, or even reduce the weight of said base plate **1301**.
(210) Each of the concaves/concave assemblies **101**, **500**, **600**, **700**, **800** can be fitted with a

mounting assembly **1500**, which can include the first over-center catch **1501**, second over-center catch **1502**, adjacent plate **1503**, and an end plate **1504**. The adjacent plate **1503** and the end plate **1504** are offset an acute angle.

(211) FIGS. **53**A-**55**D show MOG limiter assemblies **1600**R, **1600**L, **1650**R, **1650**L (**1600**, **1650**). The MOG limiter assemblies **1600**, **1650** include rub bars **1601**, **1651**, rails **1602**, **1652**, extended mounting portions **1603**R, **1603**L, **1653**R, **1653**L, and hinge points **1604**, **1654**.

(212) The MOG limiter assembly **1600**, **1650** is formed by coupling two rails **1602**, **1652** to another via a plurality of spaced rub bars **1601**, **1651** attached therebetween. The rails **1602**, **1652** are substantially parallel to one another, and the rub bars **1601**, **1651** are substantially parallel to each other. As shown, the rails **1602**, **1652** follow a substantially curvilinear path so as to conform to an outer side of a concave within a combine. Moreover, the plurality of rub bars **1601**, **1651** are spaced from another such that they will fit in between axial bars of a concave. At a first end of the MOG limiter assembly **1600**, **1650** is the hinge points **1604**, **1654**. Each of the hinge points **1604** (for the MOG limiter assembly **1600**) and the hinge points **1654** (for the MOG limiter assembly **1650**) form an end of the rails (**1602** and **1652** respectively) by tapering off so as to be easily insertable into apertures within cross bars of a concave as can be seen in later figures (see, e.g., FIGS. **56**B, **57**C).

(213) At an end opposite the hinges points **1654** are the extended mounting portions **1603**R, **1603**L, **1653**R, **1653**L. As shown, the extended mounting portions **1603**R, **1603**L, **1653**R, **1653**L can vary in shape, weight, and size, but are each configured to provide a mounting place for the over-center latch assemblies **1300**R, **1300**L, **1400**R, **1400**L. A surface of each of the extended mounting portions **1603**R, **1603**L, **1653**R, **1653**L to which the over-center latch assemblies **1300**, **1400** attach is angled to allow for connection to catches on the concave in such a way that will provide tension of the MOG limiter assemblies **1600**, **1650** against an underside of the concave. A preferred method of attaching the over-center latch assemblies **1300**, **1400** to the extended mounting portions **1603**R, **1603**L, **1653**R, **1653**L is via plug weld in the slots **1308**, **1408**.

(214) In a first example shown in FIGS. **53**A-**53**B, a right-side quarter wrap (QW) MOG limiter assembly **1600**R includes solid, extended mounting portions **1603**R that extend from the rails **1602**. The mounting portions **1603**R each have an edge intended to align with the slots **1308** in each side of the base plate **1301** of the over-center latch assembly **1300**.

(215) Similarly, in a second example shown in FIGS. **53**C-**53**D, a left-side quarter wrap (QW) MOG limiter assembly **1600**L can similarly include the solid, extended mounting portions **1603**L that extend downward from the rails **1602**. The mounting portions **1603**L each have an edge intended to align with the slots **1308** in each side of the base plate **1301** of the over-center latch assembly **1300**.

(216) In a third example shown in FIGS. **54**A-**54**D, a right-side quarter wrap (QW) MOG limiter assembly **1650**R includes solid, extended mounting portions **1653**R that extend from the rails **1652**. The mounting portions **1653**R each have an edge intended to align with the slots **1408** in each side of the base plate **1401** of the over-center latch assembly **1400**.

(217) Similarly, in a fourth example shown in FIGS. **55**A-**55**D, a left-side quarter wrap (QW) MOG limiter assembly **1650**L can similarly include the solid, extended mounting portions **1653**L that extend downward from the rails **1652**. The mounting portions **1653**L each have an edge intended to align with the slots **1408** in each side of the base plate **1401** of the over-center latch assembly **1400**.

(218) FIGS. **56**A-**57**C show a concave assembly **2600**, **2650**. The concave assembly **2600**, **2650** includes side rails **2601**, **2651**, a center rail **2602**, **2652**, cross bars **2603**, **2653**, a first end **2604**, **2654**, a second end **2605**, **2655**, and a wire lock clevis pin **2656**.

(219) The side rails **2601**, **2651** form sides of the concave assembly **2600**, **2650** each concluding at both of the first end **2604**, **2654** and the second end **2605**, **2655**. Extending down a center of the concave assembly **2600**, **2650** is the center rail **2602**, **2652** substantially parallel to each of the side rails **2601**, **2651**. The concave assembly **2600**, **2650** further includes the cross bar **2603**, **2653**

which may be a single cross bar **2603**, **2653** or two cross bars **2603**, **2653**. The side rails **2601**, **2651** may includes apertures along a length of the side rails **2601**, **2651** so as to allow for varying placement of the cross bar **2603**, **2653** depending on a size of MOG limiters or cover plates positioned therebetween. The cross bar **2603**, **2653** includes apertures therein to receive the hinge points **1604**, **1654**, **1704**, **1754** described herein. The first end **2604**, **2654** of the concave assembly **2600**, **2650** generally corresponds to the right side of the concave and the second end **2605**, **2655** generally corresponds to the left side of the concave. As shown, the second end **2605**, **2655** may include a c-clamp portion for affixing the concave assembly **2600**, **2650** within the threshing chamber of the combine. The first end **2604**, **2654** is configured to secure to the first mounting assembly **1500** and the second end **2605**, **2655** is configured to secure to the second mounting assembly **1550**.

(220) The concave assembly **2600** further includes MOG limiter assemblies **1600** positioned at an underside of the concave with MOG limiter assemblies **1600**R at a right side of the concave assembly **2600** and MOG limiter assemblies **1600**L positioned at a left side of the concave assembly **2600**. As shown, this correlates to using the over-latch assembly **1300**R and **1300**L respectively. The concave assembly **2650** further includes MOG limiter assemblies **1650** positioned at an underside of the concave with MOG limiter assemblies **1650**R at a right side of the concave assembly **2650** and MOG limiter assemblies **1650**L positioned at a left side of the concave assembly **2650**. As shown, this correlates to using the over-latch assembly **1300**R and **1300**L respectively.

(221) As shown in the concave assembly **2650** in FIG. **57**B-**57**C, the concave assembly **2650** further includes the wire lock clevis pin **2656** for further locking in place the MOG limiter assemblies **1600** to the concave. The wire lock clevis pin **2656** may be any locking mechanism suitable for extending through the center rail **2652** and the rail **1652** of the MOG limiter assembly **1600** (apertures can be seen throughout the figures of apertures existing on a portion of the MOG limiter assembly that transitions from the rail **1602**, **1652** to the extended mounting portion **1603**, **1653**, and one such visual of the pin **2656** extending through these apertures can be seen in FIG. **57**B). As can be seen in FIG. **57**C, two wire lock clevis pins **2656** are included on the left side of the concave assembly **2650** whereas only one wire lock clevis pin **2656** is included on the right side of the concave assembly **2650**. This is because of a wider stance of the rails **1652** on the MOG limiter assembly **1600**R as opposed to the narrower stance of the rails **1652** on the MOG limiter assembly **1600**L. As shown with previous embodiments, the MOG limiter assembly can be configured to alternate (see, e.g., FIG. **5**) as opposed to adopting wider and narrower stance rails **1652**. Although not shown in the figures, nothing precludes the wire lock clevis pin **2656** from being incorporated into the concave assembly **2600**.

(222) When fully assembled, the MOG limiter assemblies **1600**, **1650** will be held in place against the concave **2600** via insertion of the hinge points **1604**, **1654** into apertures of the cross bar **2603**, **2653**, and latching down of the over-center latch assemblies **1300**, **1400** via the loops **1305**, **1405** attaching to the catches **1501**, **1502**, **1551**, **1552** and the converging of the handles **1302**, **1402** towards the base plates **1301**, **1401**, and further via inclusion of the wire lock clevis pins **2656**.

(223) FIGS. **58**A-**60**D show cover plate assemblies **1700**R, **1700**L, **1750**R, **1750**L (**1700**, **1750**). The cover plate assemblies **1700**, **1750** include a curvilinear plate **1701**, **1751**, rails **1702**, **1752**, extended mounting portions **1703**R, **1703**L, **1753**R, **1753**L, hinge points **1704**, **1754**, and contact protrusions **1705**, **1755**.

(224) The cover plate assembly **1700**, **1750** is formed by coupling two rails **1702**, **1752** to another via the curvilinear plate **1701**, **1751** attached therebetween. The rails **1702**, **1752** are substantially parallel to one another. As shown, the rails **1702**, **1752** follow a substantially curvilinear path along the curvilinear plate **1701**, **1751** so as to conform to an outer side of a concave within a combine. At a first end of the MOG limiter assembly **1700**, **1750** is the hinge points **1704**, **1754**. Each of the hinge points **1704** (for the MOG limiter assembly **1700**) and the hinge points **1754** (for the MOG

limiter assembly **1750**) form an end of the rails (**1702** and **1752** respectively) by tapering off so as to be easily insertable into apertures within cross bars of a concave as can be seen in later figures (see, e.g., FIGS. **62**, **65**).

(225) At an end opposite the hinges points **1754** are the extended mounting portions **1703**R, **1703**L, **1753**R, **1753**L. As shown, the extended mounting portions **1703**R, **1703**L, **1753**R, **1753**L can vary in shape, weight, and size, but are each configured to provide a mounting place for the over-center latch assemblies **1300**R, **1300**L, **1400**R, **1400**L. The right-side cover plate assembly **1700**R, **1750**R can include a triangular mounting portion **1703**R, **1753**R that includes a cavity therethrough. Like the solid, extended mounting portion **1603**, it is characterized by an edge (the lower leg of the triangle) intended to align with the slots **1308** in each side of the body **1301** of the over-center latch assembly **1300**. A surface of each of the extended mounting portions **1703**R, **1703**L, **1753**R, **1753**L to which the over-center latch assemblies **1300**, **1400** attach is angled to allow for connection to catches on the concave in such a way that will provide tension of the cover plate assemblies **1700**, **1750** against the underside of the concave. A preferred method of attaching the over-center latch assemblies **1300**, **1400** to the extended mounting portions **1703**R, **1703**L, **1753**R, **1753**L is via plug weld in the slots **1308**, **1408**.

(226) The over-center latch assembly **1300** is also designed so as to be useable with each of the cover plate assemblies **1700**, **1750**, as shown in FIGS. **58**A-**60**D.

(227) FIGS. **61-65** show a concave assembly **2700**, **2750**. The concave assembly **2700**, **2750** includes side rails **2701**, **2751**, a center rail **2702**, **2752**, cross bars **2703**, **2753**, a first end **2704**, **2754**, a second end **2705**, **2755**, and a wire lock clevis pin **2756**.

(228) The side rails **2701**, **2751** form sides of the concave assembly **2700**, **2750** each concluding at both of the first end **2704**, **2754** and the second end **2705**, **2755**. Extending down a center of the concave assembly **2700**, **2750** is the center rail **2702**, **2752** substantially parallel to each of the side rails **2701**, **2751**. The concave assembly **2700**, **2750** further includes the cross bar **2703**, **2753** which may be a single cross bar **2703**, **2753** or two cross bars **2703**, **2753**. The side rails **2701**, **2751** may includes apertures along a length of the side rails **2701**, **2751** so as to allow for varying placement of the cross bar **2703**, **2753** depending on a size of MOG limiter or cover plates positioned therebetween. The cross bar **2703**, **2753** includes apertures therein to receive the hinge points **1604**, **1654**, **1704**, **1754** described herein. The first end **2704**, **2754** of the concave assembly **2700**, **2750** generally corresponds to the right side of the concave and the second end **2705**, **2755** generally corresponds to the left side of the concave. As shown, the second end **2705**, **2755** may include a c-clamp portion for affixing the concave assembly **2700**, **2750** within the threshing chamber of the combine. The first end **2704**, **2754** is configured to secure to the first mounting assembly **1500** and the second end **2705**, **2755** is configured to secure to the second mounting assembly **1550**.

(229) The concave assembly **2700** further includes cover plate assemblies **1700** positioned at an underside of the concave with cover plate assemblies **1700**R at a right side of the concave assembly **2700** and cover plate assemblies **1700**L positioned at a left side of the concave assembly **2700**. As shown, this correlates to using the over-latch assembly **1300**R and **1300**L respectively. The concave assembly **2750** further includes cover plate assemblies **1750** positioned at an underside of the concave with cover plate assemblies **1750**R at a right side of the concave assembly **2750** and cover plate assemblies **1750**L positioned at a left side of the concave assembly **2750**. As shown, this correlates to using the over-latch assembly **1300**R and **1300**L respectively.

(230) As shown in the concave assembly **2750** in FIG. **64-65**, the concave assembly **2750** further includes the wire lock clevis pin **2756** for further locking in place the cover plate assemblies **1700** to the concave. The wire lock clevis pin **2756** may be any locking mechanism suitable for extending through the center rail **2752** and the rail **1752** of the cover plate assembly **1700** (apertures can be seen throughout the figures of apertures existing on a portion of the cover plate assembly that transitions from the rail **1702**, **1752** to the extended mounting portion **1703**, **1753**,

and one such visual of the pin **2756** extending through these apertures can be seen in FIG. **64**). As can be seen in FIG. **65**, two wire lock clevis pins **2756** are included on the left side of the concave assembly **2750** whereas only one wire lock clevis pin **2756** is included on the right side of the concave assembly **2750**. This is because of a wider stance of the rails **1752** on the cover plate assembly **1700**R as opposed to the narrower stance of the rails **1752** on the cover plate assembly **1700**L. As shown with previous embodiments, the cover plate assembly can be configured to alternate (see, e.g., FIG. **5**) as opposed to adopting wider and narrower stance rails **1752**. Although not shown in the figures, nothing precludes the wire lock clevis pin **2756** from being incorporated into the concave assembly **2700**.

(231) When fully assembled, the cover plate assemblies **1700**, **1750** will be held in place against the concave **2700** via insertion of the hinge points **1704**, **1754** into apertures of the cross bar **2703**, **2753**, and latching down of the over-center latch assemblies **1300**, **1400** via the loops **1305**, **1405** attaching to the catches **1501**, **1502**, **1551**, **1552** and the converging of the handles **1302**, **1402** towards the base plates **1301**, **1401**, and further via inclusion of the wire lock clevis pins **2756**.

(232) When fully assembled, and as shown in FIGS. **56**A-**56**B the concave **100** can therefore attach one or more of the left-side quarter wrap (QW) MOG limiter assembly **1100** and one or more of the right-side quarter wrap (QW) MOG limiter assembly **1200** by way of the extended mounting portions **1603**, the over-center latch assembly **1300**, and the mounting assembly **1500**.

(233) When fully assembled, and as shown in FIGS. **61-62** the concave **100** can therefore attach one or more of the left-side quarter wrap (QW) cover plate assembly **900** and one or more of the right-side quarter wrap (QW) cover plate assembly **1000** by way of the extended mounting portions **1603**, the over-center latch assembly **1300**, and the mounting assembly **1500**.

(234) FIG. **66** shows the locking nut **1800** with a nylon friction ring **1801** incorporated therein comprises a nut with a threaded nylon insert. The locking nut with a nylon friction ring is shown in use for example in FIGS. **36**, **38**, **40-42**, **44-45**, **48**A-**48**B, **49**A-**50**B, **50**A-**51**C, among others. The nylon friction ring can be added instead of a locking nut for increased locking strength and can be particularly useful when in use with the screw loop so as to retain an initial position of the screw loop without concern of it rotating to either of a longer or shorter length for when in use with the assembly.

(235) It is to be appreciated that any of the concave, MOG limiter, cover plate, over-center latch, etc. described herein may be intermixed and diversified in various combinations. One such example given by way of example and not of limitation is that FIGS. **56**A-**57**C shows only MOG limiter assemblies attached to the concave, and FIGS. **61-65** show only cover plate assemblies attached to the concave and are designated as concave assembly **2600** and concave assembly **2700**. This is in no way limiting to the combinations possible for the concaves as the concave itself within each of the concave assembly **2600** and the concave assembly **2700** can be the same concave. As such, components shown in each of these assemblies may be intermixed so as to have one MOG limiter assembly **1600**R and three cover plate assemblies **1700**R, **1700**L, **1750**L within the concave assembly or two and two placed at various positions etc. The same goes for over-center latch assemblies and extended mounting portions with intermixing and all sorts of combinations are possible.

(236) It is to be appreciated any one or more of the concaves/concave assemblies **101**, **500**, **600**, **700**, **800**, mounting assembly **200**, cover plates/cover plate assemblies **300**, **400**, **900**, **1000**, and MOG limiter assemblies **105**, **1100**, **1200**, can be equipped with sensors and technologies that can automatically adjust based on field conditions and crop type.

(237) It is to be appreciated any one or more of the concaves/concave assemblies **101**, **500**, **600**, **700**, **800**, mounting assembly **200**, cover plates/cover plate assemblies **300**, **400**, **900**, **1000**, and MOG limiter assemblies **105**, **1100**, **1200**, can be included in various combinations to form various kits. For example, according to embodiment, the kit includes components selected from the group consisting of: one or more concave, preferably three or more round bar concaves; one or more pairs

of left-hand and right-hand cover plates, preferably two or more pairs; one or more pairs of left-hand and right-hand quarter-wrap mog-limiters, preferably six or more pairs; an auger bed bearing holder; a heavy duty stainless steel toggle locking latch used on one or more of the MOG limiters and/or cover plates, preferably sixteen of said latches; one or more lynch pins for securing said latches, preferably sixteen of said lynch pins; one or more square wire lock pins, preferably nine or more of said lock pins; and any combination thereof.
(238) From the foregoing, it can be seen that the present disclosure accomplishes at least all of the stated objectives.
EXAMPLES
(239) In a preferred embodiment, a number of MOG limiters **105** are employed and strategically positioned on the concaves in positions #1, #2 and #3 within the combine as reflected on FIG. **21**. This configuration provides maximum retention of the pods or other small grain within the rotary chamber. Using soybeans as an example, this increased retention at the beginning of the threshing operation allows for increased separation of the bean from the pod and also prevents the pods from dropping through to the cleaning system. In positions #2 and #3, the MOG limiters are solely utilized which allows for the threshed beans to flow out of the rotary chamber while retaining the MOG within the chamber. Experimentation has shown that this configuration is highly effective in opening the soybean pods to free the soybeans in the threshing operation and, thereafter, timely and efficiently exit the threshed soybeans from the rotary chamber to prevent damage or grinding while preventing MOG from flowing out of the chamber and into the cleaning system.
(240) The number of cover plates utilized are directly related to the difficulty in getting the soybean pods to open and release the soybeans in the threshing operation. Therefore, when the farmer is harvesting greener soybeans which have a higher moisture content, retention time within the threshing chamber is increased by the addition of cover plates in position #2 and, depending upon conditions, #3. If the soybean plants being harvested at a particular time are well dried down, then lesser retention time in the threshing chamber is required, so the use of one or more additional MOG limiters in position #1 may allow for maximal efficient soybean harvest.
(241) As a result of the improvements discussed in the present disclosure, the farmer is able to finish harvest of soybeans in an unirrigated field or a field having sandy soil with a combine set-up utilizing two or more MOG limiters in position #1, and then while in the field alter the configuration to match the configuration of FIG. **56**A when transitioning to harvest the same variety of soybean plants having the same maturity, but which are greener due to irrigation or other soil factors which afford access to more moisture to the plants or factors which result in the differential moisture content of the plants under harvest.
(242) Experimentation has identified a beneficial set up of a John Deere combine. This set up involves the use of three modified round bar concaves in positions #1, #2 and #3. MOG limiters are readily used on such concave designs and afford the farmer the maximum of adjustability based upon the harvest conditions being faced at the time. As for the Case IH combine, experimentation has identified a beneficial set up whereby the Max Flow concave is utilized on position #1 on which cover plates may be used as required and modified round bar concaves in positions #2 and #3 whereupon MOG limiters may be adjustably employed by the farmer.
(243) Regarding the embodiment shown throughout FIGS. **22-23**, the modified MOG limiting concave assembly **100** was tested with wheat, and the wheat farmer was pleased with the results utilizing a plate on both the upper section and lower section of the 1st column with MOG Limiters in the remaining positions. The modified MOG limiting concave assembly **100** utilizes a ⅝″ round bar having a ¾″ gap in between; and a ⅜″ wide MOG Limiter bar.
(244) Regarding the embodiment shown throughout FIGS. **24-25**, the modified MOG limiting concave assembly **200** utilizes alternating ⅝″ round bars with ⅜″ square bars, having a ⅛″ gap with the top of the square bar positioned at an elevation ¼″ above the top point of the round bars.
(245) Regarding the embodiment shown throughout FIGS. **26-27**, the modified MOG limiting

concave assembly **300** utilizes a ¼″ round bar having a ⅛″ gap; and a grouping of ½″ square bars having a 11/8″ gap; and further having a ⅛″ elevation of the top of the square bars above top elevation point of the round bars.

(246) Regarding the embodiment shown throughout FIGS. **28**-**29**, the modified MOG limiting concave assembly **400** utilizes ⅝″ round bar having a ¾″ gap between each; every 3rd bar is a ⅜″ square bar having a ⅞″ gap on each side in relation to the round bar; and further having a ⅛″ elevation of the top of the square bars above top elevation point of the round bars.

(247) Comparative field data was gathered in a first field for the harvest of soybeans utilizing the following assemblies: the (1) OEM-JD S670 35′ Conventional 3″ and (2) the Calmer-JD S680 40′ draper, as shown in Table 1:

(248) TABLE-US-00001 x3 Loss Sample Loose Beans/ per Area Beans Pods Pod Total Sq Ft BU/AC 35 sq ft 168 21 63 231.00 6.60 1.65 35 sq ft 114 0 0 114.00 3.26 0.81 Difference 3.34 0.84 Average BU/AC Loss with OEM ⅝″gap Concaves: 1.65 Average BU/AC Loss with Calmer ¾″gap Concaves: 0.81 Calmer BU/AC Advantage: 0.84 50.6%

(249) Comparative field data was gathered in a second field for the harvest of soybeans utilizing the following assemblies: the (1) OEM-JD 9670 30′ Head and (2) the Calmer-JD S680 40′ draper, as shown in Table 2:

(250) TABLE-US-00002 x3 Loss Sample Loose Beans/ per Area Beans Pods Pod Total Sq Ft BU/AC 30 sq ft 165 25 75 240.00 8.00 2.00 30 sq ft 113 7 21 134.00 4.47 1.12 Difference 3.53 0.88 Average BU/AC Loss with OEM ⅝″gap Concaves: 2.00 Average BU/AC Loss with Calmer ¾″gap Concaves: 1.12 Calmer BU/AC Advantage: 0.88 44.2%

(251) Comparative field data was gathered in a third field for the harvest of soybeans utilizing the following assemblies: the (1) OEM-JD 9670 30′ Head and (2) the Calmer-JD S680 40′ draper, as shown in Table 3:

(252) TABLE-US-00003 x3 Loss Sample Loose Beans/ per Area Beans Pods Pod Total Sq Ft BU/AC 30 sq ft 98 8 24 122.00 4.07 1.02 40 sq ft 130 0 0 130.00 3.25 0.81 Difference 0.82 20.1% Average BU/AC Loss with OEM ⅝″gap Concaves: 1.02 Average BU/AC Loss with Calmer ¾″gap Concaves: 0.81 Calmer BU/AC Advantage: 0.20 20.1%

(253) The average results from the three counts are shown in Table 4:

(254) TABLE-US-00004 Average Soybean Difference: 2.56 Average BU/AC Loss with OEM 5/8″gap 1.56 Concaves with no inserts: Average BU/AC Loss with Calmer 3/4″gap 0.91 Concaves with cover plates and MOG limiters: Advantage Calmer 0.64 41.2%

(255) A high performance upgrade kit that reduces grain loss during harvest was compared across corn and soybeans, as shown in Table 5:

(256) TABLE-US-00005 Grain Crop # Samples Loss BU/AC Corn 1 Sample with JD Rotary Combine 51% 0.9 Corn 2 Samples with CASE Flagship 47.1% 1.76 Soybeans 2 Samples with JD Rotary Combine 41.2% .64

(257) Another such high performance upgrade kit that reduces grain loss during harvest was compared across different types of combines, as shown in Table 6:

(258) TABLE-US-00006 CASE CASE JD S- 3 Combine Comparison Flagship Legacy Series Kernels (BU/AC) with the kits 2.90 1.57 0.73 described herein installed 1.45 0.8 0.36

(259) FIG. **45** shows soybean pods that were harvested using a high performance upgrade kit that reduces grain loss during harvest.

(260) Another comparison among residue data in a Calmer BT Super Chopper vs. JD RowMax is shown in Table 7:

(261) TABLE-US-00007 ″Small % ″Medium % ″Large % <4″″ long″ 4-8″″ long″ >8″″ long″ 6 rows of Calmer 39.1% 31.7% 29.3% 12 Blade Choppers 6 rows of John Deere 18.4% 30.7% 50.9% RowMax Residue was separated into categories; each category was weighed. Weights are presented as a % of total weight for residue produced.

LIST OF REFERENCE CHARACTERS

(262) The following table of reference characters and descriptors are not exhaustive, nor limiting, and include reasonable equivalents. If possible, elements identified by a reference character below and/or those elements which are near ubiquitous within the art can replace or supplement any element identified by another reference character.

(263) TABLE-US-00008 TABLE 8 List of Reference Characters 100 assembly 101 concave 102 upper concave connector 102A upper concave connector (opposite side) 103 lower concave connector 104 insert slot 105 MOG limiter 106 insert 107 rail support 108 mounting plate 109 notch 110 concave crossbar 111 hinge point 112 bolts 200 mounting plate 201 threaded post 202 J-hook, threaded at one end 203 hexagonal sleeve (female threaded) 204 lever 300 left-side cover 301 curvilinear plate 302 first arced support 303 second arced support 304 slot 305 mounting plate 306 bridge support 400 right-side cover 401 curvilinear plate 402 first arced support 403 second arced support 404 slot 405 mounting plate 406 bridge support 500 first embodiment of a modified concave 501 an enhanced side flow rail 502 a C-clamp end 503 first-end round bar drive 504 second end plate 505 quarter wrap center catch 506 an enhanced flow center rail 507 round bars 508 left over center catch 509 right over center catch 600 second embodiment of a modified concave 601 an enhanced side flow rail 602 a C-clamp end 603 first-end round bar drive 604 second end plate 605 quarter wrap center catch 606 an enhanced flow center rail 607 rub bars 608 round bars 609 left over center catch 610 right over center catch 700 third embodiment of a modified concave 701 an enhanced side flow rail 702 a C-clamp end 703 first-end round bar drive 704 second end plate 705 quarter wrap center catch 706 an enhanced flow center rail 707 rub bars 708 round bars 709 left over center catch 710 right over center catch 800 fourth embodiment of a modified concave 801 an enhanced side flow rail 802 a C-clamp end 803 first-end round bar drive 804 second end plate 805 quarter wrap center catch 806 rub bars 807 round bars 808 an enhanced flow center rail 809 left over center catch 810 right over center catch 900 left-side quarter wrap (QW) cover plate assembly 901 left-side quarter wrap cover plate 902 left-side quarter wrap cover plate support 903 left-side cover plate over center handle anchor plate 904 left-side quarter wrap cover plate rail 905 J-bolt 906 threaded long-steel coupling nut 907 threaded long hex head screw 908 threaded nylon insert lock nut 909 steel washer 910 CMF over center handle 1000 right-side quarter wrap (QW) cover plate assembly 1001 right-side quarter wrap cover plate 1002 right-side quarter wrap cover plate support 1003 right-side cover plate over center handle anchor plate 1004 right-side quarter wrap cover plate rail 1005 J-bolt 1006 threaded long-steel coupling nut 1007 threaded long hex head screw 1008 threaded nylon insert lock nut 1009 steel washer 1010 CMF over center handle 1100 left-side quarter wrap (QW) MOG limiter assembly 1101 left-side quarter wrap MOG limiting rail 1102 left-side MOG limiting rub bar 1103 left-side ML over center handle anchor plate 1104 J bolt 1105 threaded long steel coupling nuts 1106 threaded long hex head screw 1107 threaded nylon insert locknut 1108 steel washer 1109 CMF over center handle 1200 right-side quarter wrap (QW) MOG limiter assembly 1201 right-side quarter wrap MOG limiting rail 1202 right-side MOG limiting rub bar 1203 right-side ML over center handle anchor plate 1204 J bolt 1205 threaded long steel coupling nuts 1206 threaded long hex head screw 1207 threaded nylon insert locknut 1208 steel washer 1209 CMF over center handle 1300 center over latch 1301 base plate (e.g., pentagonal body) 1302 handle 1303 padlock hole 1304 latch arm 1305 loop 1306 center pin 1307 ear 1308 slot 1309 hole 1310 threaded nylon insert locknut 1311 subbase 1400 over-center latch assembly 1401 base plate (e.g., pentagonal body) 1402 handle 1403 locking hole 1404 lever 1405 loop 1406 center pin 1407 ear 1408 slot 1409 hole 1410 threaded nylon insert locknut 1411 subbase 1412 locking mechanism 1450 over-center latch 1500 first mounting assembly 1501 first over-center catch 1502 second over-center catch 1503 adjacent plate 1504 end plate 1550 second mounting assembly 1551 first over-center catch 1552 second over-center catch 1554 end plate 1600R MOG limiter assembly (right) 1600L MOG limiter assembly (left) 1651 rub bar 1652 rail 1653R extended mounting portion (right) 1653L extended mounting portion (left) 1654 hinge point 1700R cover plate assembly (right) 1700L cover plate assembly (left) 1701 curvilinear plate

1702 rail 1703R extended mounting portion (right) 1703L extended mounting portion (left) 1704 hinge point 1705 contact protrusion 1750R cover plate assembly (right) 1750L cover plate assembly (left) 1751 curvilinear plate 1752 rail 1753R extended mounting portion (right) 1753L extended mounting portion (left) 1754 hinge point 1755 contact protrusion 1800 locknut 1801 nylon insert 2600 concave 2601 side rail 2602 center rail 2603 cros frame 2604 first end 2605 second end 2650 concave 2651 side rail 2652 center rail 2653 cross frame 2654 first end 2655 second end 2656 wire lock clevis pin 2700 concave assembly 2701 side rail 2702 center rail 2703 cross frame 2704 first end 2705 second end 2750 concave assembly 2751 side rail 2752 center rail 2753 cross frame 2754 first end 2755 second end 2756 wire lock clevis pin

Glossary

(264) Unless defined otherwise, all technical and scientific terms used above have the same meaning as commonly understood by one of ordinary skill in the art to which embodiments of the present disclosure pertain.

(265) The terms “a,” “an,” and “the” include both singular and plural referents.

(266) The term “or” is synonymous with “and/or” and means any one member or combination of members of a particular list.

(267) As used herein, the term “exemplary” refers to an example, an instance, or an illustration, and does not indicate a most preferred embodiment unless otherwise stated.

(268) The term “about” as used herein refers to slight variations in numerical quantities with respect to any quantifiable variable. Inadvertent error can occur, for example, through use of typical measuring techniques or equipment or from differences in the manufacture, source, or purity of components.

(269) The term “substantially” refers to a great or significant extent. “Substantially” can thus refer to a plurality, majority, and/or a supermajority of said quantifiable variables, given proper context.

(270) The term “generally” encompasses both “about” and “substantially.”

(271) The term “configured” describes structure capable of performing a task or adopting a particular configuration. The term “configured” can be used interchangeably with other similar phrases, such as constructed, arranged, adapted, manufactured, and the like.

(272) Terms characterizing sequential order, a position, and/or an orientation are not limiting and are only referenced according to the views presented.

(273) The “invention” is not intended to refer to any single embodiment of the particular invention but encompass all possible embodiments as described in the specification and the claims. The “scope” of the present disclosure is defined by the appended claims, along with the full scope of equivalents to which such claims are entitled. The scope of the disclosure is further qualified as including any possible modification to any of the aspects and/or embodiments disclosed herein which would result in other embodiments, combinations, subcombinations, or the like that would be obvious to those skilled in the art.

Claims

1. An over-center latch for attaching a cover plate or MOG limiter to an outer surface of a concave for a combine comprising: a subbase including a pair of ears protruding therefrom; a handle in rotational communication with the pair of ears at a base of the handle; a center pin positioned between the base of the handle and a distal end of the handle configured to be in rotational communication with the handle; a lever attached at a base of the lever to the center pin, wherein the center pin allows the lever to rotate with respect to a handle; a loop at a distal end of the lever opposite the base of the lever; and a base plate attached to a bottom of the subbase; wherein the cover plate or MOG limiter includes: rails running curvilinear along a length of the cover plate or MOG limiter; extended mounting portions extending outwardly from an outer surface of each of the rails and each of the extended mounting portions forming an edge, wherein each of the edges

are flat and angled away from the rails such that no portion of the edges are parallel to any portion of the outer surface of the rails from which the extended mounting portions extend; wherein the base plate attaches to the edges of the extended mounting portions; wherein either the handle or the lever can be pulled into a vertical position orthogonal the subbase.

2. The over-center latch of claim 1, further comprising a lock hole protruding from the subbase in a same direction as the pair of ears.

3. The over-center latch of claim 2, further comprising a locking mechanism positioned within the lock hole.

4. The over-center latch of claim 3, wherein the locking mechanism is a linchpin.

5. The over-center latch of claim 1, further comprising a threaded portion along a length of the lever that extends into the center pin, wherein the center pin includes a threaded section to receive the threaded portion of the lever.

6. The over-center latch of claim 5, further comprising a locking nut that can attach the threaded portion of the lever to the center pin.

7. The over-center latch of claim 6, further comprising a nylon friction ring positioned within the locking nut that dampens vibration.

8. The over-center latch of claim 1, wherein the handle includes an angled portion on the distal end of the handle to assist with lifting the handle, wherein the angled portion is angled away from the subbase when the handle is positioned substantially parallel to the subbase.

9. The over-center latch assembly of claim 1, wherein the subbase includes a plurality of apertures and the base plate includes a plurality of holes, wherein the apertures may align with the holes for connecting the base plate to the subbase.

10. The over-center latch assembly of claim 9, further comprising threaded rods or screws inserted through the holes and the apertures for connecting the subbase and base plate, wherein the threaded rods or screws include a locking nut with a nylon insert to fasten the threaded rods or screws to the over-center latch assembly.

11. The over-center latch assembly of claim 1, wherein the subbase and the base are a monolithic component via weld such that the pair of ears extend directly from the base plate.

12. The over-center latch assembly of claim 1, wherein the base plate includes slots positioned at opposite ends of the base plate and the base plate attaches to the MOG limiter or the cover plate via plug welds in the slots.

13. The over-center latch assembly of claim 12, wherein the over-center latch assembly secures the cover plate or the MOG limiter to an outer surface of a concave.

14. The over-center latch assembly of claim 13, wherein the concave includes catches for the loop to connect to for securing the cover plate or the MOG limiter to the concave.

15. A method for securing a MOG limiter or cover plate to a concave via an over-center latch assembly, comprising: positioning a hinge point of the MOG limiter or cover plate within a cross bar of the concave; lifting a handle of the over-center latch such that a lever operatively connected to the handle extends outwardly away from the MOG limiter or cover plate; placing a loop that is positioned at a distal end of the lever on a catch of the concave; rotating the handle a direction opposite the lifting such that lever retracts back towards the MOG limiter or cover plate; and placing a locking mechanism through a lock hole on the over-center latch assembly after the handle has been rotated past the lock hole.

16. The method of claim 15, further comprising securing the cover plates and/or MOG limiters to the concave.

17. The method of claim 15, further comprising inserting a wire lock clevis pin through rails of the cover plates and/or MOG limiters and a center rail of the concave.

18. The method of claim 15, wherein the cover plates and/or MOG limiters comprise half a length of the concave.

19. A combine including a concave, comprising: a plurality of cover plates and/or MOG limiters

positioned on an outer surface of the concave; and a plurality of over-center latch assemblies configured to secure the plurality of cover plates and/or MOG limiters to the outer surface of the concave; wherein each of the cover plates and/or MOG limiters includes one of the plurality of over-center latch assemblies secured thereto; wherein each of the plurality of over-center latch assemblies comprises a base plate with a pair of ears extending therefrom; a handle operatively connected to the pair of ears; a center pin positioned along a length of the handle and capable of rotating with relation to the handle; a lever extending from the center pin; and a loop positioned at a distal end of the lever; wherein the cover plates and/or MOG limiters are secured to the concave via insertion of hinge points into a cross bar of the concave, wherein the hinge points are positioned on the cover plate and/or MOG limiter at an end opposite the over-center latch assembly, wherein the cover plate and/or MOG limiter is further secured to the concave via attaching the loops of the over-center latch assemblies to catches positioned at each end of the concave.